Journal of Hydrology 632 (2024) 130955

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Groundwater–surface water exchange from temperature time series: A comparative study of heat tracer methods

Erik Saphores [a], Sarah Leray [a,b], Francisco Suárez [a,c,d,e,*]

[a] Departamento de Ingeniería Hidráulica y Ambiental, Pontificia Universidad Católica de Chile, Santiago, Chile
[b] Centro de Cambio Global (CCG), Pontificia Universidad Católica de Chile, Santiago, Chile
[c] Centro de Desarrollo Urbano Sustentable (CEDEUS), Santiago, Chile
[d] Centro de Excelencia en Geotermia de los Andes (CEGA), Santiago, Chile
[e] Centro del Desierto de Atacama (CDA), Santiago, Chile

ARTICLE INFO

This manuscript was handled by Corrado Corradini, Editor-in-Chief, with the assistance of William Payton Gardner, Associate Editor.

Keywords:
Silala River
International watercourse
Thermal methods
Groundwater–surface water interactions

ABSTRACT

The importance of the interaction between groundwater and surface water is increasingly being recognized for both understanding and managing water systems. Many efforts have been made to characterize and quantify groundwater–surface water exchange. In particular, temperature–based methods have quickly established themselves given their monetary and practical advantages. In the last 15 years, several methods for interpreting passive temperature time series measured in the streambed have been developed. Still, the benchmarking of these methods has only been carried out in specific and distinct hydrological conditions.

This article aims to fill this research gap by benchmarking the performance of six commonly used methods for deriving seepage fluxes using a two-year-long temperature time series covering various meteorological and thermal conditions. This work compares three analytical methods that calculate seepage flux using the amplitude and/or phase of the temperature signals, and three numerical methods that use different schemes to inversely solve the one–dimensional heat transport equation in the streambed. The temperature measurements were made in the context of an international dispute between Chile and Bolivia over the status and use of the waters of the Silala River, Northern Chile. Flux estimations are tested against Darcy's flux derived from measured hydraulic gradient and hydraulic conductivity.

Flux estimations from the benchmarked methods ranged from −0.5 to 3.5 m/d (with positive fluxes directed downwards), whereas fluxes estimated using Darcy's law ranged from 0.5 to 6 m/d. Results show that the amplitude method is the best–performing method. This method is best suited for estimating the direction of the fluxes, while the method using both the thermal amplitude and phase is best suited for monthly flux trends, and the combination of a Local Polynomial (LP) method and a Maximum Likelihood Estimator (MLE) method (LPMLEn) is appropriate for estimating flux in transient conditions.

The use of heat as a tracer proved to be an effective tool for monitoring groundwater–surface water exchange in a river reach for two years, and yielded exchange flux estimates with lower point-scale variability than Darcy's law.

1. Introduction

The connection between groundwater and surface waters has been recognized globally (Scanlon et al., 2023). Yet, while the processes and water compartments participating in hydrological cycles have been well established, the quantitative estimation of the compartments' volume and fluxes (in particular between surface water and groundwater) remains limited (Brooks et al., 2015; Gleeson et al., 2021; McCallum et al., 2012; Rosenberry et al., 2021). Considering surface water and groundwater as part of a continuum is especially opportune given the high and increasing demand for water resources, which support not only industry (Barnett et al., 2005; Lettenmaier et al., 1999) and agriculture (Arnell and Liu, 2001; Panagoulia and Dimou, 1996) but also natural habitats (Bower et al., 2004; Sophocleous, 2004). Therefore, since integrated groundwater/surface water analyses reduces uncertainty in water resources management (Duque and Rosenberry, 2022; Ren et al., 2023), it

* Corresponding author at: Departamento de Ingeniería Hidráulica y Ambiental, Pontificia Universidad Católica de Chile, Santiago, Chile.
E-mail address: fsuarez@uc.cl (F. Suárez).

https://doi.org/10.1016/j.jhydrol.2024.130955
Received 7 July 2023; Received in revised form 13 December 2023; Accepted 13 February 2024
Available online 28 February 2024
0022-1694/

is crucial to have available different methods that can estimate groundwater–surface water exchange (Constantz et al., 2003; Kurylyk et al., 2019), as well as to accurately monitor the relevant variables that govern groundwater–surface water interaction (Hatch et al., 2006).

Flow exchange between surface water and groundwater has been extensively studied using traditional methods such as seepage meters, differential gauging, hydrograph separation, environmental tracer methods, and approaches based on Darcy's law (Cascarano et al., 2020; Cook et al., 2018; Humphrey et al., 2022; Kalbus et al., 2006; Klaus and McDonnell, 2013; Ma et al., 2022; Ruehl et al., 2006). Over the last 15 years, heat-based methods have become popular techniques to characterize surface water–groundwater interaction (Anderson, 2005; Briggs et al., 2018; Constantz, 2008; Irvine et al., 2017; Naranjo et al., 2023; Saar, 2011; Simon et al., 2022) due to their many advantages, such as being a natural environmental tracer, and their ease of measurement, low cost and non-intrusive instrumentation, which facilitate systematic deployment (Hatch et al., 2006; Saar, 2011).

Both passive and active heat-based methods rely on the basic principle that a flowing fluid can transfer heat from one location to another; hence, tracing temperature in natural hydrosystems may help in determining the exchange between water bodies of different temperatures and origins. Daily temperature fluctuations of river water/streambed interface can be compared to those of the underlying media to deduce the magnitude and direction of the surface water–groundwater exchange (in this work, we use the convention that positive fluxes are directed downwards). Accordingly, various analytical and numerical methods based on the analysis of two or more temperature time series acquired at different depths in the streambed have been proposed to characterize surface water–groundwater interactions. Although all these methods solve the heat transport equation, they can produce relative differences in the seepage flux magnitude estimation (Luce et al., 2013; McAliley et al., 2022; McCallum et al., 2012; Irvine et al., 2015a). These differences can be due to violations in the assumption of one-dimensional vertical flow, (e.g., due to lateral or hyporheic flow), or directly from specific procedures in the temperature time-series analysis (e.g., Fourier transform (Luce and Tarboton, 2010), harmonic regression (Taylor et al., 2007), and recursive estimation (McAliley et al., 2022)), and from the variable of interest (e.g., amplitude ratio (Hatch et al., 2006; Keery et al., 2007), phase shift (Hatch et al., 2006; Keery et al., 2007) or both (Luce et al., 2013; McCallum et al., 2012)). Furthermore, the differences in seepage flux estimation from different methods in the same study site is typically used to improve confidence in the results (Duque and Rosenberry, 2022; Ren et al., 2023).

Validation and intercomparison of heat tracer methods is complicated by the fact that a great number of studies have been developed and used on relatively short temperature time series (generally less than three months) that do not cover a full hydrological year (Briggs et al., 2014; Cardenas, 2010; Cuthbert and Mackay, 2013; Gordon et al., 2012; Hatch et al., 2006; Healy and Ronan, 1996; Irvine et al.,2015a,b; Koch et al., 2015; Lautz, 2010; Luce et al., 2013; Ren et al., 2023; Rosenberry et al., 2016; Schneidewind et al., 2016; Suárez et al., 2024a; Vandersteen et al., 2015; van Kampen et al., 2022; Shi et al., 2023). The performance of these methods, populated with short-term data, is appropriate when large vertical downward gradients exist, but it is limited when seasonal variations in hydrological conditions occur (e.g., when thermal gradients are reversed or where surface temperatures are more stable than subsurface temperatures). Moreover, the thermal parameters involved in heat transport are generally estimated from the literature rather than by direct measurement, consequently increasing the uncertainty of flux estimations (Glose et al., 2019, 2021; Gordon et al., 2012; Hatch et al., 2006; Healy and Ronan, 1996; Irvine et al., 2015b; Keery et al., 2007; McAliley et al., 2022; McCallum et al., 2012; Munz et al., 2011; Munz and Schmidt, 2017; Schneidewind et al., 2016; Simon et al., 2022; Vandersteen et al., 2015; van Kampen et al., 2022). Additionally, some thermal methods have been validated against simulated temperature time series from synthetic models or from laboratory and controlled experiments, which do not allow the evaluation of the methods in real field conditions (Cardenas, 2010; Cuthbert and Mackay, 2013; Glose et al., 2019, 2021; Hatch et al., 2006; Irvine et al., 2015a, 2015b; Lautz, 2010; Munz et al., 2011; Ren et al., 2023). Finally, some authors have questioned the ability of one-dimensional heat-based methods to estimate seepage flux magnitudes (Ghysels et al., 2021), especially in low-seepage flux environments and in upwelling conditions (Briggs et al., 2014; Glose et al., 2019, 2021).

However, despite the significant instrumental development and the diversity of techniques, few studies have attempted to extensively benchmark the calculations obtained from the panel of available heat-based methods. Accordingly, the objective of this work is to evaluate the ability of existing analytical and numerical methods to produce hydraulically consistent seepage flux estimates under different meteorological and thermal conditions. Six methods are tested against a 22-month-long temperature time series collected in the Silala River, Northern Chile, as part of a monitoring campaign in the context of a legal dispute between Chile and Bolivia (Aravena et al., 2024; Gómez et al., 2024; Lagos et al., 2024; Latorre and Frugone-Álvarez, 2024; Muñoz et al., 2024; Suárez et al., 2024a; Taylor and Peach, 2024; Wheater et al., 2024; CM, 2017). In this dispute, Chile requested the International Court of Justice to declare the Silala River as an international watercourse. This requires demonstration that it is a system comprised of both surface waters and groundwaters, forming, by their physical interconnectedness, a unitary whole flowing into a common terminus. Therefore, it was important to illustrate that groundwater and surface waters interact along the river reach, i.e., to establish that the flow between these systems is non-zero (Suárez et al., 2024a). Beyond the specific case of the Silala River dispute, localizing surface water-groundwater interactions and determining their direction can be informative enough in various applications such as hydrogeological modeling, seasonal studies at the catchment scale, biogeochemical hotspots, and environmental studies, while comparatively less complicated to implement in the field (Simon et al., 2022, 2023). During the measurement period, the study site was influenced by rainfall, snowfall and snowmelt, freezing and thawing processes, pumping tests, and discharge from an artesian well that was initially located upstream and was then moved downstream later in the measurement period, making the study site a natural laboratory for the evaluation of the methods' performances under different hydrologic, thermal, and hydraulic conditions. The results obtained from the benchmarked methods are compared to flux estimates based on Darcy's law, which uses hydraulic gradients and conductivities to determine seepage fluxes. Since head data offer a broad spatial integration compared to point measurements, Darcy's flux results in the most trusted seepage estimates at the local scale. Therefore, Darcy's law was used to validate the direction and magnitude trend of seepage fluxes inferred by thermal methods. The hydraulic gradient and thermal and hydraulic properties were measured in the field.

In the following section, we present the heat and fluid transport equation that is solved by the heat tracer methods, as well as a brief overview of each approach. Subsequently, we define each heat tracer method and Darcy's law as a traditional technique to obtain an alternative seepage flux estimate. After defining the methods, we present the study site along with the data collection scheme. Then, the results are presented, and the methods based on different indicators are evaluated and discussed. Lastly, conclusions and future perspectives are described.

2. Benchmark methodology

2.1. Theoretical background of heat tracer methods

Heat tracer methods solve the one-dimensional heat transport equation across the streambed sediments (Stallman, 1965):

$$\frac{\partial T}{\partial t} = k_e \frac{\partial^2 T}{\partial z^2} - q_z \frac{C_w}{C} \frac{\partial T}{\partial z} \tag{1}$$

where z is the depth in the streambed (m); T is the temperature (°C); t is time (s); k_e is the effective thermal diffusivity of the saturated medium ($m^2 s^{-1}$); q_z is the vertical Darcy flux, also known as seepage flux ($m\ s^{-1}$, positive downwards); and C_w and C are the volumetric heat capacities of water and the saturated medium, respectively ($J\ m^{-3} K^{-1}$). The effective thermal diffusivity in Equation (1) is given by (Hatch et al., 2006):

$$k_e = \frac{\lambda_0}{C} + \beta |v_f| = \frac{\lambda_0}{C} + \beta \left| \frac{q_z}{\eta_e} \right| \tag{2}$$

where λ_0 is the baseline thermal conductivity (i.e., without fluid flow) of the saturated medium ($J\ s^{-1} m^{-1} K^{-1}$); β is the thermal dispersivity (m); v_f is the linear velocity ($m\ s^{-1}$); and η_e is the effective porosity (-). Equation (1) considers heat transfer by advection, conduction, and dispersion, and assumes one-dimensional flow along the vertical axis. Dispersion is considered through the parameter β. The temperature time series within the streambed sediments generally exhibits the temporal evolution shown in Fig. 1, in which the amplitudes (A) of the thermal signals dampen with depth and their phase ($\Delta\varphi$) lags as depth increases.

2.2. Benchmarked analytical methods

Analytical methods consider the temperature time series of the sediments as sinusoidal signals and compare the temperature evolution at different depths. The methods of Hatch et al. (2006) and Keery et al. (2007) independently use the amplitude or phase of these signals to obtain the magnitude and direction of the exchange flux. Meanwhile, the methods of McCallum et al. (2012) and Luce et al. (2013) simultaneously use the amplitude and phase of the signals to solve Eq. (1).

Analytical methods that have been developed to determine the seepage flux (q) by solving Eq. (1) are based on the following assumptions: (i) homogeneous properties in the saturated medium; (ii) no mean geothermal gradient; (iii) thermal equilibrium between the solid and fluid phases; and (iv) thermal properties independent of temperature (Irvine et al., 2017). Furthermore, these methods assume the first-type (Dirichlet) top boundary condition, that is, an imposed temperature at the river–sediment interface, to be a sinusoidal signal and a semi-infinite domain.

2.2.1. Amplitude methods

Hatch et al. (2006) and Keery et al. (2007) developed analytical solutions by analyzing the amplitude of the temperature time series. Both solutions calculate the seepage flux using the relative amplitude, A_r, between two temperature signals at different depths:

$$A_r = \frac{A_d}{A_s} \tag{3}$$

where A_d and A_s are the amplitudes of the temperature signals (°C) of the deep and shallow sensors from a sensor pair, respectively (Fig. 1). The solution of Hatch et al. (2006) is as follows:

$$q = \frac{C}{C_w}\left(\frac{2k_e}{\Delta z} ln ln(A_r) + \sqrt{\frac{\alpha + v_t^2}{2}}\right) \tag{4}$$

with:

$$\alpha = \sqrt{v_t^4 + \left(\frac{8\pi k_e}{P}\right)^2} \tag{5}$$

where Δz is the spacing between the sensor pair (m), v_t is the thermal front velocity ($m\ s^{-1}$), and P is the period of the temperature signal (s) (typically one day).

The Keery et al. (2007) solution differs from that of Hatch et al. (2006) since it does not consider thermal dispersion. Consequently, the Hatch et al.'s (2006) solution was chosen for the amplitude method due to its ability to include dispersion effects.

2.2.2. Phase methods

Hatch et al. (2006) and Keery et al. (2007) also developed analytical solutions for the seepage flux using the phase shift, $\Delta\varphi$, between two temperature signals at different depths:

$$\Delta\phi = \Delta\phi_d - \Delta\phi_s \tag{6}$$

where $\Delta\varphi_d$ and $\Delta\varphi_s$ are the phases of the temperature signals of the deep

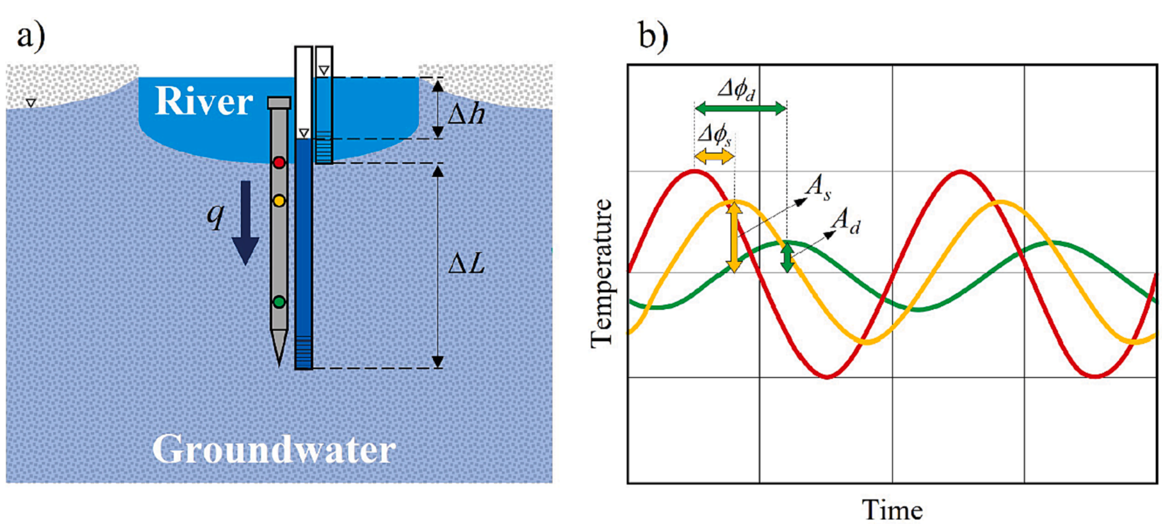

Fig. 1. Diagrams detailing the acquisition of streambed temperature data. (a) Experimental setup for temperature data collection: a buried hollow tube with temperature sensors along its length (temperature rod). The locations of the temperature sensors are shown by the red, yellow, and green circles. Two piezometers are placed beside the temperature rod, one to measure the groundwater head and the other to measure the river stage. The spacing between the screening of the piezometers (ΔL) and the hydraulic head difference (Δh) used for hydraulic gradient computations are depicted. (b) Theoretical temporal evolution of the temperature time series at different depths within the streambed sediments. With depth, the amplitude of the signals dampens, and their phase increasingly lags. For a sensor pair, the amplitude of the deeper (A_d) and shallower (A_s) sensors and the phase lag of the deeper ($\Delta\varphi_d$) and shallower ($\Delta\varphi_s$) sensor are shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and shallow sensors from a sensor pair (radians), respectively (Fig. 1). The solution of Hatch et al. (2006) is given by:

$$|q| = \frac{C}{C_w}\sqrt{\alpha - 2\left(\frac{4\pi\Delta\phi k_e}{P\Delta z}\right)^2} \tag{7}$$

The solution of Keery et al. (2007) differs from that shown in Equation (7) since it does not consider thermal dispersion, and thus the solution of Hatch et al. (2006) was chosen for the phase method due to its capability of including dispersion effects.

2.2.3. Combined amplitude and phase methods

McCallum et al. (2012) and Luce et al. (2013) developed solutions to Equation (1) by simultaneously combining A_r and $\Delta\varphi$ between two temperature signals at different depths. These two methods give the same estimates for q (Irvine et al., 2017). A relevant difference from the other heat-based methods is that thermal properties are not needed to determine seepage fluxes. Moreover, these methods also generate a time series of the estimated k_e. As the two methods yield the same results, here we only present the solution of McCallum et al. (2012):

$$q = \frac{C}{C_w}\left(\frac{\Delta z\left(P^2(A_r) - 4\pi^2\Delta\phi^2\right)}{\Delta\phi\sqrt{\left(16\pi^4\Delta\phi^4 + 8P^2\pi^2\Delta\phi^2(A_r) + P^4(A_r)\right)}}\right) \tag{8}$$

2.2.4. Seepage flux estimation

The seepage flux estimates for the three analytical methods tested here were obtained using the VFLUX program (Gordon et al., 2012; Irvine et al., 2015b): this is a toolbox developed in the MATLAB environment for the processing of temperature time series which uses Dynamic Harmonic Regression (DHR) (Young et al., 1999) to fit the phase and/or amplitude of the sinusoidal series to the temperature time series and eventually estimate the seepage flux (among other features).

These analytical methods only use two sensors to produce estimates, so a specific sensor pair had to be chosen. The decision of which sensor pair to use was based on the estimated sensor pair spacing proposed by Luce et al. (2013), and prioritizing the sensors installed near the water–sediment interface.

2.3. Benchmarked numerical methods

Several numerical approaches have been developed to estimate q from Equation (1) using a variety of algorithms, which allow using complex non-sinusoidal boundary conditions. The methods of Healy and Ronan (1996) and Munz and Schmidt (2017) numerically solve the heat transport equation using a finite difference scheme. Meanwhile, the method of McAliley et al. (2022) formulates the heat transport equation as a state-space model and uses recursive estimation to obtain the fluxes by applying the extended Kalman filter and Rauch–Tung–Striebel smoother algorithms. The method of van Kampen et al. (2022) uses a transfer function approach to obtain the exchange fluxes using a maximum-likelihood estimator. The method of van Kampen et al. (2022) solves the heat transport equation in the frequency domain, considering frequencies over the entire spectrum defined by the user (van Kampen et al., 2022). This approach is promising, as Wörman et al. (2012) showed that considering multiple frequencies in these analyses can improve the estimation of exchange fluxes and thermal diffusivities.

2.3.1. Recursive estimation

The recursive estimation method was developed by McAliley et al. (2022). It formulates Equation (1) as a state-space model, where the spatial derivatives are computed using finite differences. It follows a two-step procedure: prediction and correction. The Extended Kalman Filter algorithm (Särkkä, 2013) is used at every time step for which temperature data are available to predict and correct the estimates, advancing through the time series. Then, the Extended Rauch–Tung–Striebel smoother (Särkkä, 2013) is used for smoothing, using all available information of estimates in a backward pass through the measurements.

For the application of the recursive estimation method (McAliley et al., 2022), the *tempest1d* Python library was used. The optimal discharge process variance was 2.36×10^{-8} m^2 d^{-2}, which was obtained via the L-Curve criterion (Hansen, 1992).

2.3.2. FLUX-BOT

The FLUX-BOT method was developed by Munz and Schmidt (2017). FLUX-BOT is a software that applies a centered Crank–Nicolson implicit finite-difference scheme and finds the q value that minimizes the sum of squared differences between the measured and modeled temperatures. This estimate of q is constant over the evaluation period. To obtain a flux value that varies in time, FLUX-BOT uses a moving time window to analyze fragments of the temperature time series, obtaining an estimate for q and then moving the window forward in time.

For the application of the FLUX-BOT method (Munz and Schmidt, 2017), the namesake MATLAB(R) program, with a 96-h sliding window, was used for flux calculations.

2.3.3. Lpmlen method

The LPMLEn method was developed by van Kampen et al. (2022) based on the work of Vandersteen et al. (2015) and Schneidewind et al. (2016). It works in the frequency domain and combines a Local Polynomial (LP) method with a Maximum Likelihood Estimator (MLE) to estimate q using n temperature sensors. The Fast Fourier Transform is used to convert the temperature time series to the frequency domain, and the LP method is then used to reduce spectral leakage and estimate noise levels. Subsequently, an MLE is used to estimate q and the thermal properties of the medium (if they were not specified) using the information of n temperature sensors and considering multiple frequencies.

Two domains can be used to estimate q: a semi-infinite domain and a bounded-domain model. Both domains use the time series of the shallowest temperature sensor as a Dirichlet boundary condition. The difference between models is the choice of the boundary condition on the deeper end of the domain: the semi-infinite domain model uses a second-type (Neumann) boundary condition, specifying that the temperature gradient approaches zero with increasing depth; and the bounded-domain model uses the time series of the deepest temperature sensor as a Dirichlet boundary condition.

For the application of the LPMLEn method, the LPMLEn MATLAB codes were used with a sliding window of 10 days. For the seepage flux calculations, the semi-infinite domain model was chosen.

2.4. Reference method based on Darcy's law

One of the most widely used methods for quantifying groundwater–stream water exchange is Darcy's law:

$$q = K\frac{\Delta h}{\Delta L} \tag{9}$$

where K is the saturated hydraulic conductivity of the medium (m s^{-1}), Δh is the hydraulic head difference between surface water and groundwater (m), and ΔL is the spacing between the piezometers used to determine the hydraulic gradient, $\Delta h/\Delta L$ (Baxter et al., 2003; Lee and Cherry, 1979; Ma et al., 2022) (Fig. 1).

The hydraulic conductivity of the medium can be measured with a permeameter. However, this parameter can vary over several orders of magnitude for a given sediment texture, meaning that the estimates of q produced by Darcy's law using the measured hydraulic conductivity can be highly uncertain (Constantz, 2008; Niswonger and Prudic, 2003; Smith and Schwartz, 1981). Moreover, due to physical and biological processes this parameter may display a variable behavior (Rosenberry et al., 2021).

3. Study site and field data collection

The study site is located in the Silala River, in Northern Chile (Wheater et al., 2024). The Silala River has an average flow of ~170 l/s at the Chile-Bolivia international border (Muñoz et al., 2024), and widths that vary from ~0.4 to more than 3 m, but typically on the order of ~1 m (Mao, 2024). In 1999, this river was the subject of a legal dispute between Chile and Bolivia over the status and use of its waters, with Bolivia claiming its waters as exclusively Bolivian (CM, Vol. 2017, pp. 11–15). Due to Bolivia's contention to recognize the Silala River as a transboundary watercourse, Chile requested that the International Court of Justice declare that the Silala river system is an international watercourse governed by customary international law (CM, Vol. 2017). In this context, several studies were performed to demonstrate that the Silala river system is an international watercourse (Gómez et al., 2024; Suárez et al., 2017; Alcayaga, 2017; Arcadis, 2017; Herrera and Aravena, 2017; Latorre and Frugone, 2017), as defined by Rieu-Clarke et al. (2012,p. 75).

Temperature data were gathered by the installation of temperature rods (TRs). Initially, five TRs were installed in November 2016 along a ~2 km reach downstream of the Chile–Bolivia border (Suárez et al., 2024a) (Study Area 1 in Fig. 2). In November 2017, five additional TRs were deployed between Quebrada Negra and Inacaliri (Study Area 2 in Fig. 2). Two piezometers were installed in each TR to measure the hydraulic gradient between the river and the fluvial aquifer beneath it. This paper focuses on TR5 since it experienced the largest number of different transient conditions, which is beneficial for the assessment of heat tracing methods. One of these transient conditions was the change in the discharge point of an artesian well, named SPW-DQN, which is located beside the Silala River (Fig. 2) (Gómez et al., 2024). This well discharges warm waters from a deeper confined aquifer. Before January 15th, 2018, the SPW-DQN well discharged its waters to the river and the aquifer beneath it by leaking approximately 30 m upstream of TR5, thus influencing both the thermal and hydraulic measurements at this location. From January 15th, 2018, the well's outflow was discharged directly to the river through a pipe approximately 50 m downstream of TR5.

The design of the TRs is based on that proposed by Naranjo and Turcotte (2015). Each one has a hollow steel rod (tube with a length of 1.5 m and a diameter of 0.0254 m [1 in.]), with temperature loggers (DS1922L iButtons, Maxim Integrated, San José, CA; accuracy: $\pm$ 0.5 °C; resolution: $\pm$ 0.0625 °C) located at different depths below the river–sediment interface. The temperature sensors are integrated into an electrical circuit board that allows the sensor position to be rearranged based on field requirements without extracting them and enables data collection without moving the loggers. The temperature records used in this research comprise the period between February 1st, 2017, and November 13th, 2018. Temperature data were collected at a frequency of 1 h over the whole period.

The depths at which temperatures were measured at TR5 were varied during the above-mentioned period. During the first part of the measurement period (Period A)—from February 1st, 2017, to April 1st, 2018—sensors were placed at depths of 0, 0.254, 0.303, 0.354, 0.452, 0.552, 0.652, and 0.851 m. During the second part of the measurement

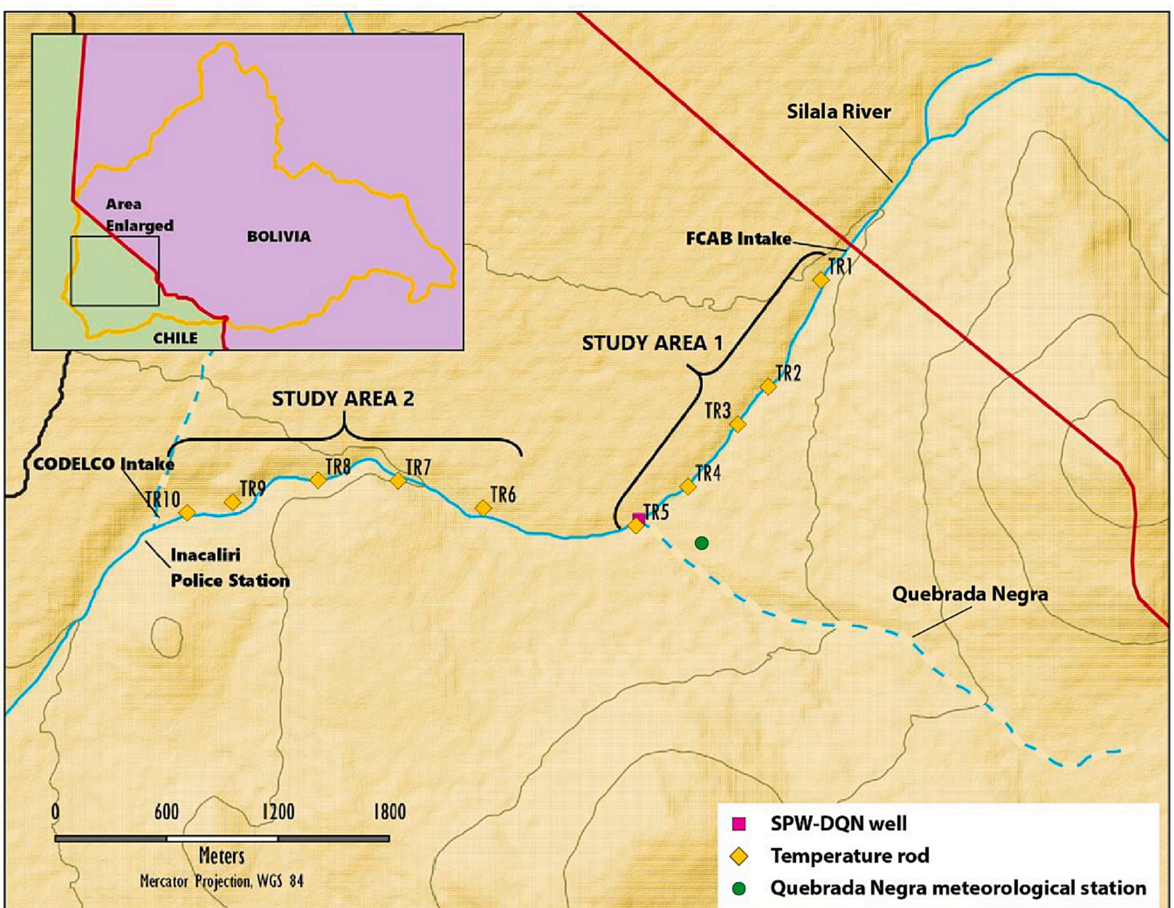

Fig. 2. Map of the study site showing the location of the temperature rods (TRs), the location of the SPW-DQN well, and the location of the Quebrada Negra meteorological station. At the location of each TR, there is a pair of piezometers measuring groundwater and surface water levels. The TRs were operational from November 2016 (Study Area 1) and from November 2017 (Study Area 2). All the TRs were removed and dismantled in June 2019. The light blue segmented lines correspond to ephemeral streams. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

period (Period B)—from April 6th, 2018, to November 13th, 2018—sensors were placed at depths of 0.02, 0.07, 0.17, 0.27, 0.37, and 0.47 m. The sensor deployment depths were changed because the temperatures recorded by the deeper sensors were practically invariant at a daily scale. For the analytical methods, a sensor pair had to be chosen to obtain seepage flux estimates since they only use two sensors. The sensor pair whose estimated spacing was closest to their actual spacing (Figs. S1 and S2 in the Supplementary Material) was assumed to provide the most trustworthy estimates. During Period A, the sensors at depths of 0.354 and 0.452 m were used, whereas during Period B, the sensors at depths of 0.27 and 0.37 m were selected.

Thermal conductivity and volumetric heat capacity of the streambed sediments were measured by means of a thermal properties' analyzer (KD2 Pro, Decagon Devices, Pullman, WA), which determines these properties using the dual probe method with transient heating (Carslaw and Jaeger, 1959; Bristow et al., 1994). Thermal conductivity and volumetric heat capacity were measured four times at each TR site, within a radius of ~1 m from the TR. The thermal conductivity was 1.940 ± 1.452 W m^{-1} K^{-1} (mean ± standard deviation) and the volumetric heat capacity was 3.613 ± 1.268 MJ m^{-3} K^{-1} (Suárez et al., 2024a). To avoid loss of generality, the thermal dispersivity (β) was set to 0.001 m, as suggested by Hatch et al. (2006) for the spatial scales used in this study, even though it has been shown to have a negligible effect on groundwater–surface water exchange (Keery et al., 2007; Munz et al., 2011; Rau et al., 2012).

The water level in the aquifer beneath the river–sediment interface was measured in one of the piezometers. The piezometers have a diameter of 0.0381 m (1.5 in.), a length of 1.5 m, and a pressure transducer (HOBO Water Level (4 m) Data Logger, Onset Computer Corporation, Bourne, MA; typical error: 4 mm; maximum error: 8 mm; resolution: 1.4 mm) installed inside. The river stage was measured using a similar pressure transducer installed inside a fully perforated PVC pipe (0.04 m diameter, 0.5 m length). The data collected from the pressure transducers were compensated for barometric variations. Water level data were collected at a frequency of 15 min over the whole period. The vertical distance between the river-sediment interface and the piezometer screening was 1.03 m, and was used to estimate the vertical hydraulic gradients.

Hydraulic conductivity estimates were obtained by testing undisturbed samples of the sediments with a variable-head permeameter (KSAT, UMS, Munich, Germany) (Johnson et al., 2005). The samples were 8 cm in diameter and 5 cm in height. Three measurements of saturated hydraulic conductivity yielded a value of 14.3 ± 1.38 m/d (mean ± standard deviation) (Suárez et al., 2024c). The porosity of the medium was not measured; therefore, we assumed a porosity of 0.38 as recommended by Briggs et al. (2016).

The correct detection of estimated seepage flux direction and monthly seepage flux trends was described by true positive (TP) and true negative (TN) classes, respectively; whereas incorrect detection was analyzed using false positive (FP) and false negative (FN) classes. With these classes, accuracy (Γ) can be estimated as (Yáñez-Morroni et al., 2023):

$$\Gamma = \frac{TP + TN}{TP + TN + FP + FN} \tag{10}$$

Precipitation and snow cover data were analyzed to examine their correlation with the temporal evolution of the temperatures measured at TR5, and with the surface water and groundwater levels. Precipitation was measured at the Quebrada Negra meteorological station (Muñoz et al., 2017) (Fig. 2). This station has a rain gauge, which registered daily precipitation, among other meteorological variables. Snow cover data are related to snowmelt, the latter of which in turn lowers the temperature measurements at TR5 as it mixes with the river. Snow cover data were obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) (500-m resolution) (Hall and Riggs, 2016). The Google Earth Engine (GEE) platform was employed for image treatment (Gorelick et al., 2017). The Normalized Difference Snow Index (NDSI) (Zhang et al., 2019) was used to distinguish between snow cover and clouds; a threshold of 0.3 was employed for this procedure, since Yañez-Morroni et al. (2024) found that it represented well the actual snow cover observations in the basin. The areas considered for the snow cover analysis are shown in the Supplementary Material (Fig. S3), and were defined as the areas that could contribute snowmelt to the Silala river and Quebrada Negra based on their proximity to the respective watercourses.

4. Results

4.1. Hydraulic gradient

The recorded river stage and groundwater level are displayed in Fig. 3, along with daily precipitation for the entire measurement period.

The river stage and groundwater levels are relatively stable until January 2018. Precipitation produces a considerable rise in groundwater levels. Precipitation also affects the river stage, but this impact is not as large and long-lasting as its effects on groundwater levels. The large field-saturated hydraulic conductivity values of the fluvial deposits in which the aquifer is housed (0.2–6.2 m/day; Gómez et al., 2024), the relatively small size of this fluvial aquifer (Blanco and Polanco, 2024; Peach and Taylor, 2024; Taylor and Peach, 2024), and the large baseflow index of the Silala River (~0.92; Muñoz et al., 2024) support that water table variations in the fluvial aquifer due to precipitation events are more responsive than streamflow. The abrupt changes in water levels observed on January 15th, 2018, were due to a pumping test performed at the SPW-DQN well, whereas the rapid decrease in the river stage on January 15th, 2018, is due to the change of the discharge point of the SPW-DQN well; on the same day, the groundwater level at TR5 also experienced a decline since deep groundwater no longer leaked into the shallow fluvial aquifer. The influence of the SPW-DQN artesian well before and after its discharge point was changed is depicted in Fig. 4. As shown at the right side of Fig. 4a), a groundwater mounding occurred as the water of the artesian flowed near the terrain surface into the river. After the discharge point of the SPW-DQN artesian well was moved downstream TR5, this mounding disappeared (right side of Fig. 4b)). This hydraulic behavior was inferred from the groundwater levels measured in both SPW-DQN (DIFROL–DICTUC 2018) and TR5 (Suárez et al., 2017).

On January 23rd, 2018, the well valve was gradually closed over a period of four hours, and it was left closed until January 28th, 2018, causing an increase in groundwater level. Then, between January 28th and 31st, 2018, the valve was gradually opened until it was completely open. This provoked a decrease in the groundwater level. Finally, on February 23rd, 2018, the well cap was removed, exposing it to atmospheric pressure (DIFROL-DICTUC, 2018).

Based on the measurements of surface water and groundwater levels at TR5, the hydraulic gradient was calculated for the entire measurement period, and seepage fluxes were subsequently calculated using Darcy's law; these fluxes are presented later in this paper. The hydraulic gradient for the duration of head measurements suggested persistent losing conditions.

4.2. Temperature and snow cover time series

The temperature time series measured at TR5 during Period A are displayed in Fig. 5a. Strong diurnal cycles are observed. The extreme daily minima that were observed on some days between February and April 2017 are due to precipitation events. The abrupt decrease in temperatures that was observed in June 2017 was due to a snowfall event in the Silala river basin (Fig. 5b), and the marked regime change that occurred in January 2018 was caused by the change in the discharge point of the SPW-DQN well. The snow that fell during the

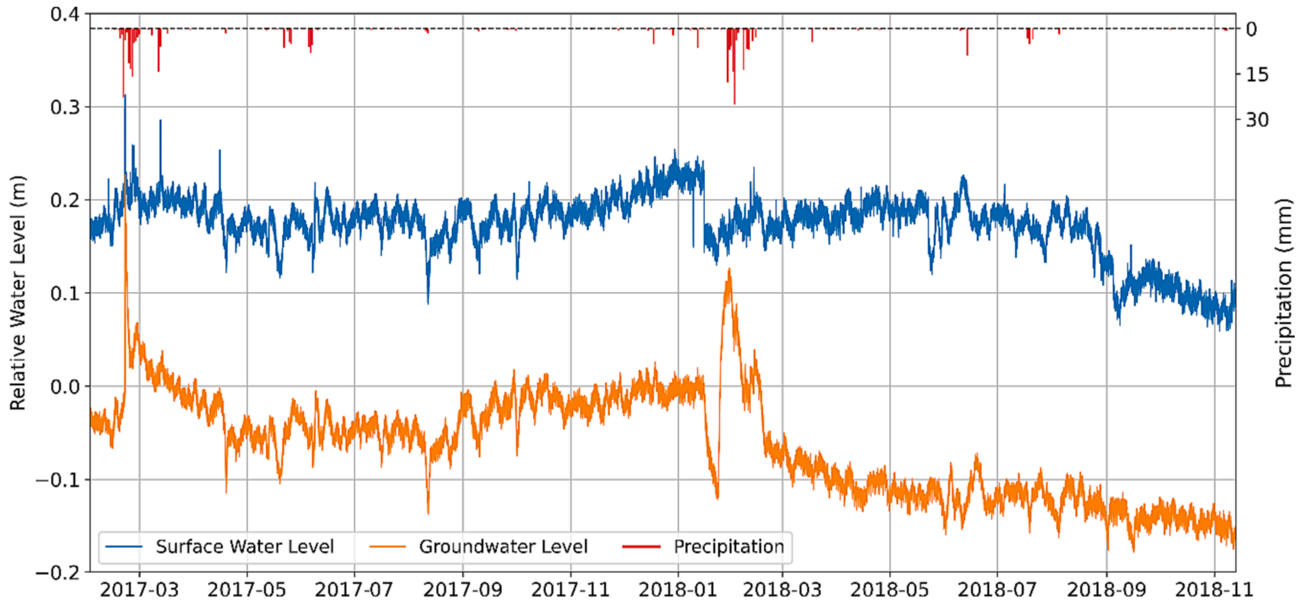

Fig. 3. River stage, groundwater levels, and precipitation in the study zone. The reported groundwater levels are relative to the groundwater level measured at 0:00 on February 1st, 2017 (the first day of record).

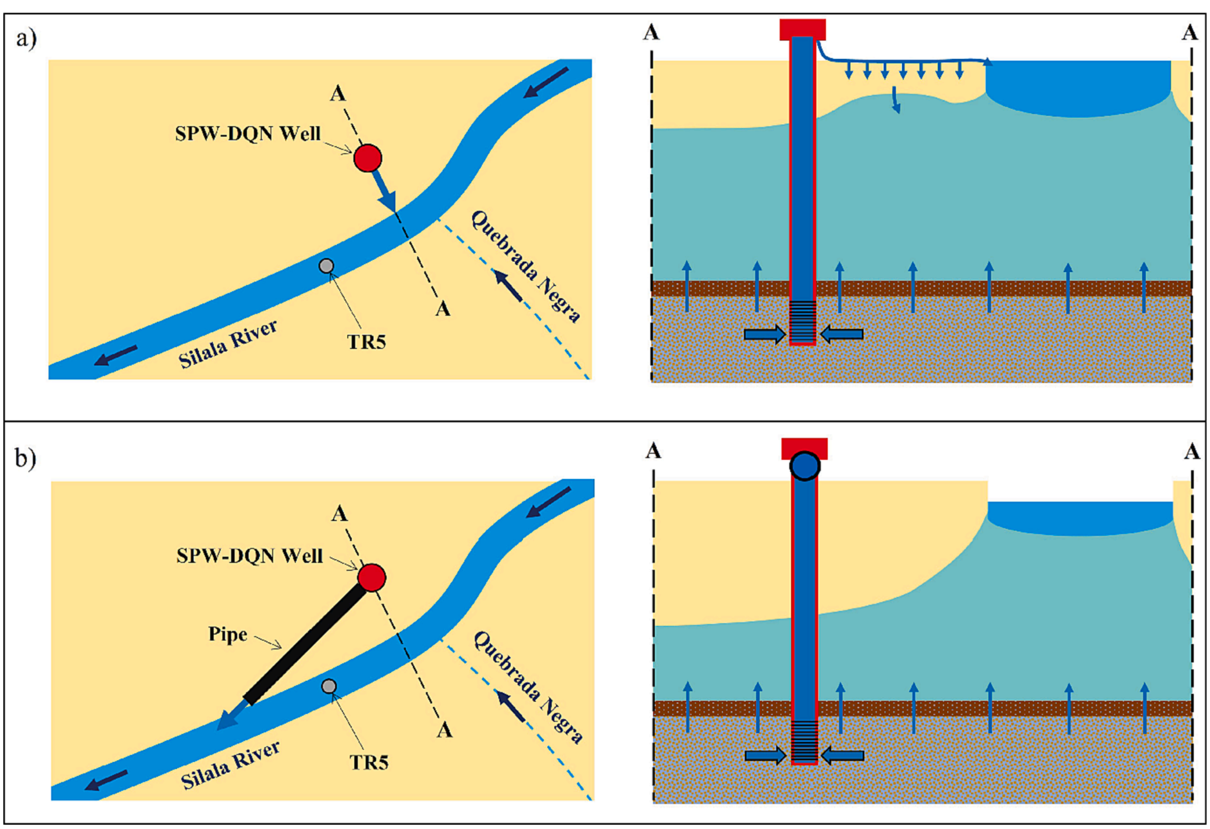

Fig. 4. Conceptual model of the influence of the SPW-DQN artesian well on the hydraulic functioning of the river and aquifer. The Quebrada Negra stream is shown as a dashed line since it is ephemeral. The hydrogeological system is composed, from top to bottom, by an unconfined fluvial aquifer, an aquitard, and a confined aquifer. (a) Conditions before January 15th, 2018. A groundwater mounding occurs due to the well's leakage infiltration as the water travels from the upper part of the well into the river. (b) Conditions after January 15th, 2018, where the discharge point of the artesian well is located downstream of TR5. In this situation, the river stage and groundwater level decrease.

summer of 2017/2018 lasted only a few days as snow cover, whereas the snow that fell during the winter of 2018 lasted nearly two months (Fig. 5b).

Before January 15th, 2018, the surface water at TR5 was a mixture of the water from the Silala River and the discharge from the artesian well. The waters of the artesian well come from a deep confined aquifer, and the well's temperature is much higher than that of the surface water flowing from Bolivia to Chile. The temperatures recorded in this period

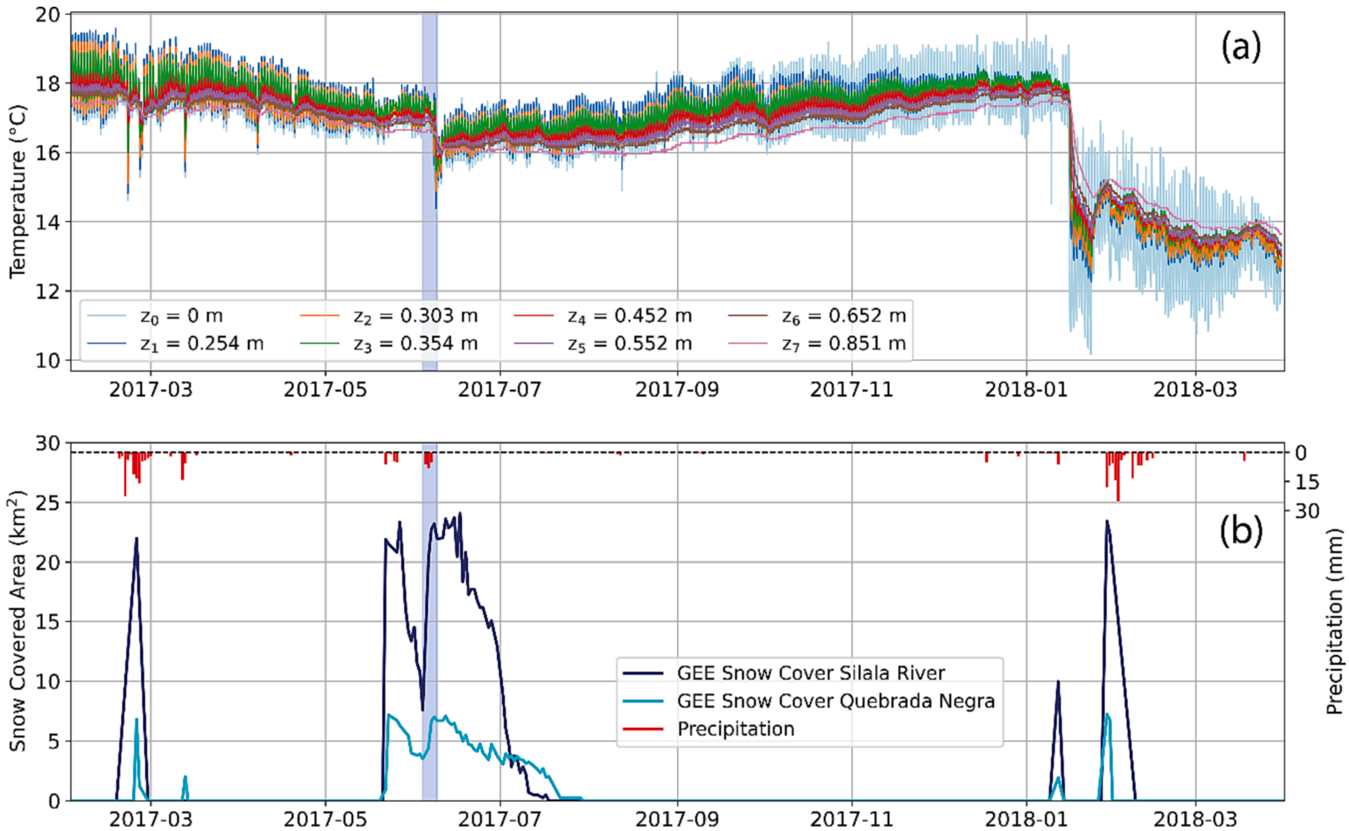

Fig. 5. Measurements of temperature, snow covered area, and precipitation made during Period A (February 1st, 2017, to April 1st, 2018): (a) Temperature time series across the streambed, where z_0 corresponds to the water–sediment interface; (b) Snow covered area in the Silala River and Quebrada Negra contributing areas (Fig. S3 in the Supplementary Material), in km^2, and precipitation at the Quebrada Negra meteorological station. The blue shaded area in the plots corresponds to the snowfall event that occurred between June 4th and 8th, 2017. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

typically ranged between 16 and 19 °C (Suárez et al., 2024a).

After January 15th, 2018, the surface water at TR5 was no longer influenced by the discharge from the SPW-DQN well. The temperatures registered during this period (i.e., between January 15th and April 1st, 2018) mostly ranged from 11 to 16 °C.

The temperature time series measured by TR5 during Period B is shown in Fig. 6a. Strong diurnal cycles were observed throughout this period. A few precipitation events were detected, which typically correspond to snowfall events (Fig. 6b). Suárez et al. (2024d) found that there were occasions when there was snow in the Silala River basin, while the pluviometers did not report precipitation. Before August 29th, 2018, streambed temperatures ranged from 8 to 15 °C and had stable amplitudes. After August 29th, 2018, the amplitude of the streambed temperature signals measured by the shallower sensors began to increase abruptly. Since every sensor on TR5 showed a similar diurnal cycle, and that is very improbable that all the sensors failed at the same time, we considered that these temperature measurements are correct. Thus, these observations were maintained in the analysis. The streambed temperatures ranged from 2 to 17 °C during September 2018, and from 10 to 17 °C from October 2018 onwards. From October 2018 onwards, the cyclic diurnal behavior remained but was stronger compared to the aforementioned period. Considering that during September 2018 the observed river temperatures were below freezing point (in Fig. 6a, z0 = 0 m corresponds to the water–sediment interface), that snowmelt occurred before September (Fig. 6b), that the river stage had the lowest record (Fig. 3), and that the pronounced cyclic diurnal patterns, it is probable that freeze–thaw processes occurred at the water surface and/or at the water/TR interface. These processes likely contributed to the substantial daily thermal amplitude observed in the streambed during this month (Fig. 6a).

4.3. *Temperature-derived fluxes*

To demonstrate how uncertainty in the thermal and hydraulic properties affects the estimates of the different methods, seepage fluxes were obtained by varying the thermal diffusivity and the hydraulic conductivity: hydraulic conductivity was varied by two standard deviations with respect to the mean measured value, whereas thermal diffusivity was arbitrarily varied by ± 20 % with respect to the average measured value. The seepage flux estimates obtained with the different methods and the hydraulic gradient for periods A and B are shown in Figs. 7 and 8, respectively.

According to the hydraulic gradient observed in Period A, seepage fluxes should be positive (directed downwards) and nearly constant until January 15th, 2018, when the magnitudes of the seepage fluxes reduce significantly and then gradually rise to higher magnitudes than those observed before the pumping test. A strong snowfall event took place in June 2017, which produced erratic q estimates on that date for most of the methods.

During Period A, the recursive estimation method and the amplitude method demonstrated the highest accuracy in predicting flux direction (100 % and 97.0 %, respectively). This accuracy was computed by comparing flux direction estimated with the thermal methods (Fig. 7(a)–(f)) and that estimated with the hydraulic gradients (Fig. 7(h)). The snowfall event of June 2017 induced erratic estimates for the three analytical methods and the FLUX-BOT method. The LPMLEn method was the best fit for the hydraulic gradient after the pumping tests (Fig. S4 in the Supplementary Material). The seepage flux estimates of this

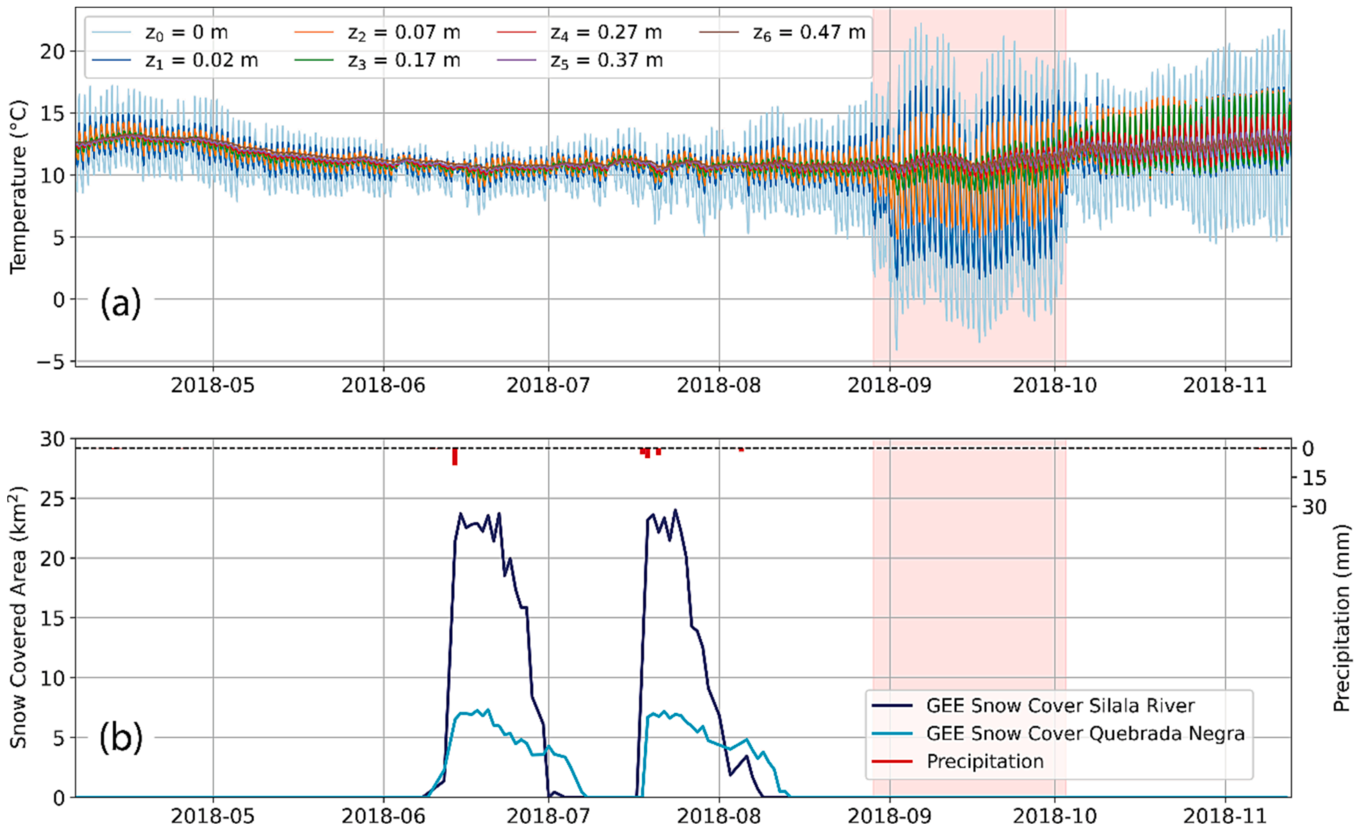

Fig. 6. Measurements made during Period B (April 6th, 2018, to November 13th, 2018): (a) Temperature time series across the streambed, where z_0 corresponds to the water–sediment interface. (b) Snow covered area in the Silala River and Quebrada Negra contributing areas (Fig. S3 in the Supplementary Material), in km^2, and precipitation at the Quebrada Negra meteorological station. The red shaded area in the plots corresponds to the period when high-amplitude thermal variations were measured. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

method reduced in magnitude at the time of the pumping tests and subsequently increased in magnitude, in agreement with the hydraulic gradient.

According to the hydraulic gradient in Period B, seepage fluxes should be positive (directed downwards) and relatively stable, and should reduce in magnitude gradually throughout the period. For Period B, the methods that best predicted the direction of the fluxes were the amplitude method and the combined amplitude and phase method (99.4 % and 84.0 % accuracy, respectively). The period during which larger temperature amplitudes were measured starting on August 29th, 2018, was associated with negative seepage flux estimates (upward fluxes) in the three numerical methods for slightly more than a month. The combined phase and amplitude method also exhibited this behavior, but only for a few days in September-October 2018. The amplitude method was the only one that correctly predicted positive fluxes over the whole increased temperature amplitudes event. From October 2018 onwards, the estimates of the numerical methods became positive again, but the recursive estimation and the FLUX-BOT methods predicted magnitudes much higher than before the event, which are not consistent with the measured hydraulic gradient. The estimates of the LPMLEn method from October 2018 onwards were similar to the estimates before September 2018 and are consistent with the hydraulic gradient. The estimates of the combined amplitude and phase method did not reach the levels that were estimated before September 2018, and the magnitude of the phase method's estimates decreased. The predictions of the amplitude method were consistent with the hydraulic gradient.

To determine whether the flux estimates follow the same trend as the hydraulic gradient, temperature-derived flux estimates from periods A and B were integrated at a monthly timescale. These monthly data were rescaled using the min–max normalization (Singh and Singh, 2020) based on the estimates of each method. The same procedure was conducted with the hydraulic gradient data. Fig. 9 shows the relationship between the temporal changes in the estimates of the normalized seepage flux and the normalized measured hydraulic gradient for every monthly time step, as well as confusion matrices that allow visualizing the performance of each method (Fawcett, 2006; Yáñez-Morroni et al., 2023).

Indicators were built to assess the performance of each thermal method by comparing their seepage flux estimates to the measured hydraulic gradient. These indicators were used to evaluate the ability of each method to extract flux direction and flux trend, and their capability to do so during highly transient conditions, such as during snowfall and pumping tests. The overall consistency of the different estimates of q with the hydraulic gradient is summarized in Table 1.

5. Discussion

Since head data offer a broader spatial integration compared to point measurements, the following discussion focuses on the ability of the different methods to estimate the direction and trend of seepage fluxes. This assessment is based on comparing the estimated values (near the water-streambed interface) with the measured hydraulic gradients, considering that seepage fluxes were not directly measured. Hence, although our work provides an intercomparison among thermal models, a limitation of this study is that the performance of these models compared to actual seepage flows is not known. It would have been possible to compare the estimated seepage fluxes with fluxes determined by more sophisticated numerical models, such as VS2DI (Hsieh et al., 2000) or COMSOL (Chen et al., 2018). Nonetheless, this is out of the scope of this work due to the inherent uncertainty related to streambed

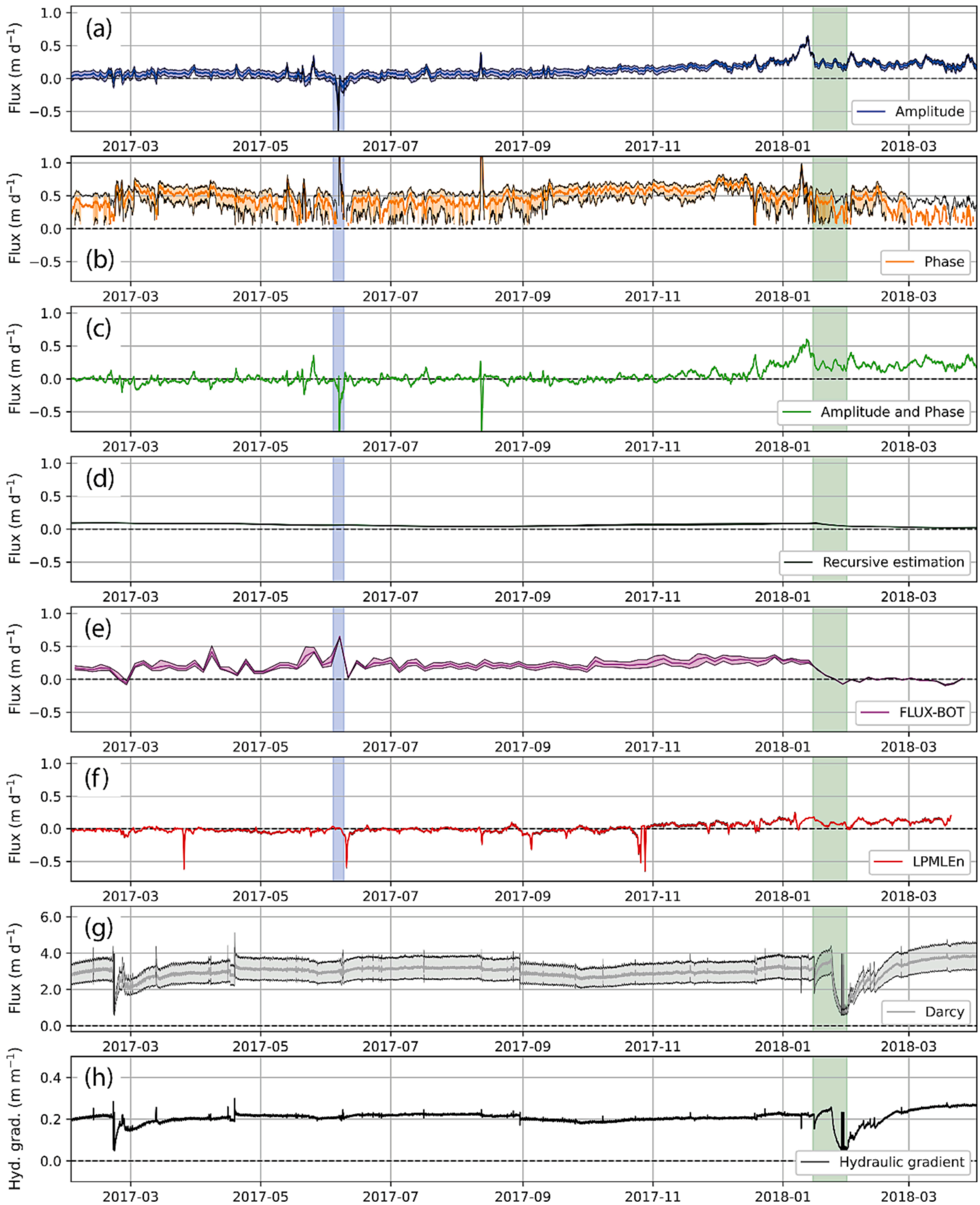

Fig. 7. Seepage flux estimates for Period A obtained using the different methods evaluated in this study (a–g), and the hydraulic gradient measured at the study site (h). The colored lines in the flux plots are the estimates obtained by using the mean measured thermal diffusivity and hydraulic conductivity. The black lines in the flux plots are the estimates obtained by varying the thermal diffusivity by ± 20 % with respect to the average measured value, and varying the hydraulic conductivity by two standard deviations higher and lower with respect to the mean measured value, respectively. The flux range in the Darcy fluxes plot (g) is significantly larger than the ranges in the other plots. Plot (f) does not include flux ranges since the associated method is not dependent on thermal parameters. The blue shaded areas in the flux plots correspond to the snowfall event that occurred between June 4th and 8th, 2017. The green shaded areas in the plots correspond to the pumping tests performed in the SPW-DQN well between January 15th and 31st, 2018. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

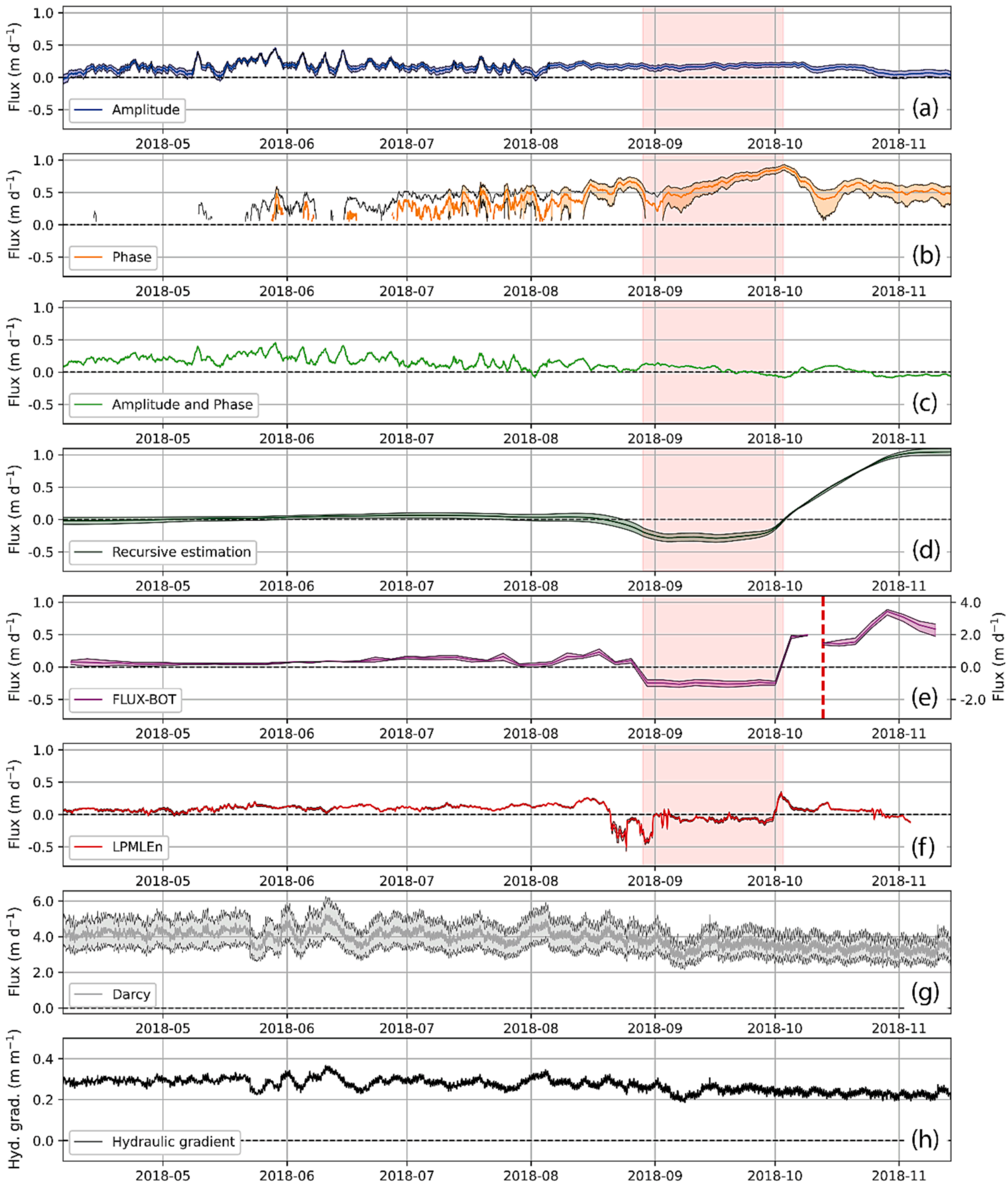

Fig. 8. Seepage flux estimates for Period B using the different methods evaluated in this study (a–g), and the hydraulic gradient measured at the study site (h). The colored lines in the flux plots are the estimates obtained by using the mean measured thermal diffusivity and hydraulic conductivity. The black lines in the flux plots are the estimates obtained by varying the thermal diffusivity by ±20 % with respect to the average measured value, and varying the hydraulic conductivity by two standard deviations higher and lower than the average measured value, respectively. The flux range covered by the Darcy fluxes plot (g) is significantly larger than the ranges in the other plots. The red dashed line indicates a scale change in the y-axis in the flux plot. Plot (f) does not include flux ranges since the associated method is not dependent on thermal parameters. The red shaded areas in the flux plots correspond to the period where high-amplitude thermal variations were measured. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

layering and/or hyporheic flow of the study site, which may result in two- or three-dimensional flow fields (Chen et al., 2015).

Even so, our comparison criteria allow to notice that, from the temperature-derived fluxes, the amplitude method (Hatch et al., 2006) best fits the direction of the hydraulic gradient. The phase method (Hatch et al., 2006) can predict flux magnitude but not its direction. The magnitudes predicted by the phase method are of the same order of magnitude as those predicted by the amplitude method, but its trend of

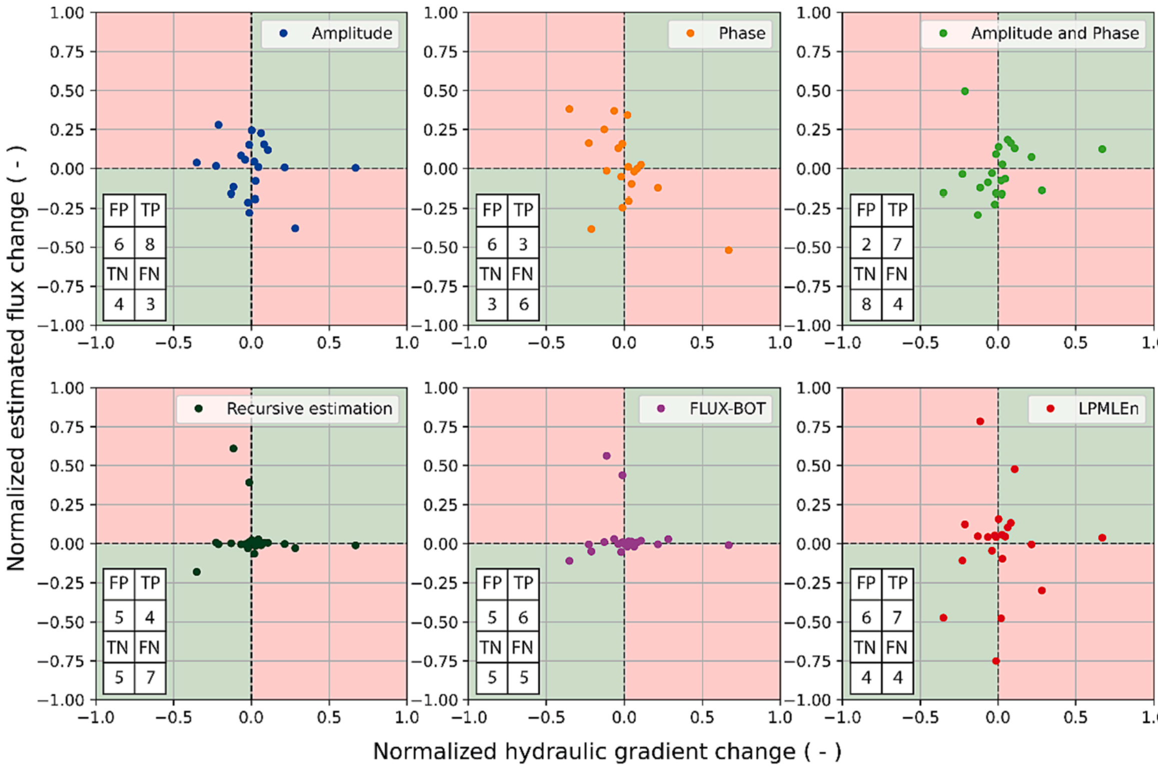

Fig. 9. Temporal changes in the estimates of the normalized seepage flux and the normalized hydraulic gradient at monthly time steps for the whole measurement period. The temporal changes are based on the value of the previous month. The points in green-shaded quadrants represent correctly estimated monthly trends–true positive (TP) and true negative (TN) values–, and the points in red-shaded quadrants represent incorrectly estimated monthly trends –false positive (FP) and false negative (FN) values–. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Consistency between seepage flux estimates and the measured hydraulic gradient. The direction estimates indicator obtained by the phase method is not included since the method only provides the flux magnitude. The indicators related to the pumping test were evaluated by analyzing Fig. S4 (Supplementary Material). The 'Reduction in magnitude with pumping test' indicator evaluates if the estimate within the pumping test period was lower in magnitude than the immediately preceding estimate. The 'Increase in flow magnitude after pumping test' evaluates if the estimates after the pumping test period were monotonically increasing, since the hydraulic gradient increased after the tests.

Method	Correct direction estimates (%)	Monthly trend extraction performance (%)	Unaltered by snowfall event in Period A	Reduction in magnitude with pumping test	Increase in flow magnitude after pumping test	Unaltered by the increased amplitudes event in Period B
Amplitude	97.8	57.1	×	✓	×	✓
Phase	–	33.3	×	✓	×	✓
Combined Amplitude and Phase	67.8	71.4	×	✓	×	✓
Recursive estimation	89.2	42.9	✓	✓	×	×
FLUX-BOT	87.5	52.4	×	✓	×	×
LPMLEn	56.8	52.4	✓	✓	✓	×

magnitude is less consistent with the changes in hydraulic gradient. The combined amplitude and phase method (McCallum et al., 2012) produces similar estimates to the amplitude method but with higher magnitudes in some periods and lower magnitudes in others, outperforming it in terms of monthly trend extraction. The disagreement between the phase and combined amplitude and phase methods, and the amplitude method, is most likely produced by the poor identification of $\Delta\varphi$. The latter may be due to signal processing errors in $\Delta\varphi$ extraction or to the 'thermal skin effect', that is, the response time delay between thermal changes in the sediments and conduction to a sensor within the TR (Cardenas, 2010; Irvine et al., 2017). The VFLUX program performs signal extraction with a DHR algorithm, a method for the analysis of non-stationary time series that is useful for extracting harmonic signals from environmental systems that exhibit transient conditions (Gordon et al., 2012; Young et al., 1999). However, this processing technique is not as prone to error when identifying the amplitude of a signal as when identifying its phase (Irvine et al., 2017), which accounts for the better compliance of the estimates obtained by the amplitude method.

The amplitude and the recursive methods have the highest predictive accuracy for flux direction (97.8 and 89.2 %, respectively). When forecasting monthly trends, the combined amplitude and phase method had an accuracy of 71.4 %, followed by the amplitude method that reached an accuracy of 57.1 %. Thus, the amplitude method outperformed the other methods overall. Although its trend-predicting ability is moderate, it was the second-highest of all the tested methods, and its flux-direction predicting ability was outstanding compared to the other methods. Our findings indicate that the method selected by Suárez et al. (2017) in their study of river–aquifer interactions in the Silala River reach (i.e., the amplitude method) was appropriate. Rosenberry et al. (2016) obtained a nearly 1:1 relationship between the seepage flux estimates obtained with the amplitude method (Hatch et al., 2006) and seepage meter measurements when saturated thermal diffusivity was evaluated in-situ, indicating that the flux magnitudes obtained in the present study are probably representative of the actual fluxes.

In this study, seepage fluxes calculated using Darcy's law had much higher magnitudes and larger variability compared to temperature-based methods. By varying the thermal diffusivity and hydraulic conductivity, temperature-derived fluxes displayed a variability of 0 to 0.5 m/d, depending on the method used, whereas fluxes based on Darcýs law are on the order of 2 m/d. Besides, for a given sediment texture, hydraulic conductivity values can vary over several orders of magnitude, whereas thermal properties are more tightly constrained, reducing the uncertainty of the flux estimates (Constantz, 2008; Duque et al., 2016; Saar, 2011). Furthermore, Darcy's law is only valid under certain ranges of hydraulic gradient (Ma et al., 2022), and therefore the shape of the actual seepage flux time series may be different from that obtained using the hydraulic gradient and Darcy's law.

Suárez et al. (2017) investigated the uncertainty of the seepage fluxes estimated with the Hatch et al. (2006) amplitude method using Monte Carlo simulations at TR1-TR5, and only accounted for expected uncertainty in streambed thermal properties, i.e.., without including uncertainty that comes from violations of model assumptions. Their 3-months analysis (November 2016-January 2017) indicates slightly increased point-scale variability (within a similar order of magnitude) compared to the current study findings. While their work demonstrated comparable uncertainty levels among the five TR's, an analysis across different river reaches using weirs revealed discrepancies between inferred losing or gaining conditions and the temperature-based seepage fluxes or Darcy's law (Suárez et al., 2017). This discrepancy is attributed to additional groundwater inputs originating from an alluvial aquifer above the Silala River (Aravena et al., 2024; Blanco and Polanco, 2024; Gómez et al., 2024; Peach and Taylor, 2024; Suárez et al., 2024b; Taylor and Peach, 2024). Hence, it is crucial to note that temperature-based seepage fluxes represent streambed conditions rather than the entire reach.

In our case study, the recursive estimation method did not display a clear ability to capture rapidly changing flow as McAliley et al. (2022) found it to be. During the pumping tests, its predicted flux decreased when the hydraulic gradient did, however, after the pumping tests, the hydraulic gradient increased to even higher levels than before the tests and the flux magnitude predictions did not increase. Also, the bounded-domain model for the LPMLEn method has been shown by van Kampen et al. (2022) to produce more trustworthy seepage flux estimations than the semi-infinite domain model; however, in our case study, the bounded-domain model produced erratic and unrealistic flux estimates when high-amplitude thermal variations occurred during Period B (e.g., −13 to 8 m/d on September 25th and 26th, 2018).

The thermal methods presented in this study do not consider two- or three-dimensional effects since these methods solve Equation (1), which assumes heat flux only in the vertical direction. Seepage flux may have non-vertical components, however, in the center of a stream, the flux is predominantly vertical (Cuthbert and Mackay, 2013; Shanafield et al., 2010). Even if the non-vertical components were relevant, the one-dimensional heat transport estimates are still valid (Cuthbert and Mackay, 2013). Moreover, the methods evaluated in this study do not consider heterogeneity, which could affect the estimates in specific cases (Irvine et al., 2015a).

Direct measurements of seepage flux were not made in this study. Thus, no assertion can be made about the absolute magnitude of seepage fluxes. The use of an automated seepage meter (Humphrey et al., 2022) combined with temperature measurements from a TR could allow this kind of analysis.

Based on hydraulic gradient measurements, it is concluded that no upwelling occurred during the measurement period. It would be desirable to assess the studied methods under upwelling conditions since they have been shown to be problematic when it comes to the use of temperature time series for estimating seepage flux (Briggs et al., 2014).

The measurement period was 22 months, which allowed the evaluation of the ability of the studied thermal methods to predict seepage flux direction and seepage flux trend, their ability to produce hydraulically consistent estimates during transient events, and their suitability to monitor seepage fluxes over long periods of time. Based on this analysis, researchers can choose the thermal method that best suits the goal of their investigation (e.g., determining whether a river reach is a gaining or losing reach).

6. Conclusions

This study is the first to compare several thermal methods for deriving seepage flux estimates based on two-years streambed temperature time series. Six thermal methods were evaluated, and the results show that the amplitude method developed by Hatch et al. (2006) outperformed the other methods. The temperature time series used to derive the flux estimates includes events of snowfall, rainfall, pumping tests, and exaggerated oscillations in the amplitudes of the thermal signals most likely due to freeze–thaw processes in the river, making it well–suited to assess transient hydraulic and thermal conditions. Darcy fluxes were obtained with piezometer and hydraulic conductivity measurements, resulting in seepage fluxes with higher magnitudes and uncertainty than those obtained with heat-based methods.

As direct measurements of seepage flux were not made, no conclusions can be made about the absolute magnitude of the estimates; only the direction and trend of the fluxes are assessed by comparing them to hydraulic gradient measurements. Nonetheless, as the data used in this work were collected in the context of a legal dispute in which Chile requested the International Court of Justice to declare the Silala River as an international watercourse, it was important to demonstrate that the seepage flux is non-zero. Identifying localized interactions between groundwater and surface waters is also important for biological, geological, geochemical, and hydrogeological reasons described elsewhere (Simon et al., 2022, 2023).

Finally, the impact of this study is that the installation of a pair of thermal sensors coupled with the use of the Hatch et al. (2006) analytical solution is an effective approach allowing a first estimate of the interaction conditions between groundwater and surface water, which can be used for the calibration of hydrological and hydrogeological models as well as for the design of more advanced experimental setups.

Future research should test the studied methods with automated seepage meter measurements to validate the magnitude of the flux estimates obtained with the different methods, as well as to further quantify the spatiotemporal uncertainty in seepage flux estimations. Additionally, it is necessary to evaluate these methods in upwelling (upward flow) scenarios since they have been shown to be problematic when it comes to the use of temperature time series for estimating seepage flux. Lastly, the heterogeneity of the sediments beneath the river was not considered in this research, and it could be an important factor when deriving seepage fluxes from temperature time series.

CRediT authorship contribution statement

Erik Saphores: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing, Conceptualization, Validation. **Sarah Leray:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Francisco Suárez:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors thank the National Dirección de Fronteras y Líites (DIFROL) de Chile for funding the projects in which these data were collected. The authors also thank the Agencia Nacional de Investigación y Desarrollo (ANID) for funding FONDECYT/1210221 and ATE/230006. F. Suárez recognizes support from the Centro de Desarrollo Urbano Sustentable (CEDEUS – ANID/FONDAP/ 1522A0002) and the Centro de Excelencia en Geotermia de los Andes (CEGA – ANID/FONDAP/15200001).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2024.130955.

References

Alcayaga, H., 2017. Characterization of the Drainage Patterns and River Network of the Silala River and Preliminary Assessment of the Vegetation Dynamics Using Remote Sensing. Memorial of the Republic of Chile, I.C.J. Pleadings, Dispute over the Status and Use of the Waters of the Silala (Chile v. Bolivia). Vol. 4, Annex I.

Anderson, M.P., 2005. Heat as a ground water tracer. Groundwater 43 (6), 951–968. https://doi.org/10.1111/j.1745-6584.2005.00052.x.

Arcadis, 2017. Detailed Hydrogeological Study of the Silala River. Memorial of the Republic of Chile, I.C.J. Pleadings, Dispute over the Status and Use of the Waters of the Silala (Chile v. Bolivia). Vol. 4, Annex II.

Aravena, R., Herrera, C., Urrutia, J., 2024. Hydrochemical and isotopic evaluation of groundwater and river water in the transboundary Silala River watershed. WIREs Water 11 (1), e1679. https://doi.org/10.1002/wat2.1679.

Arnell, N.W., Liu, C., 2001. Hydrology and water resources. In: McCarthy, J.J., Canziani, O.F., Leary, N.A., Dokken, D.J., White, K.S. (Eds.), Climate Change 2001: Impacts, Adaptation, and Vulnerability: Contribution of Working Group II to the Third Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge Univ. Press, New York, pp. 194–227.

Barnett, T.P., Adam, J.C., Lettenmaier, D.P., 2005. Potential impacts of a warming climate on water availability in snow-dominated regions. Nature 438 (7066), 303–309. https://doi.org/10.1038/nature04141.

Baxter, C., Hauer, F.R., Woessner, W.W., 2003. Measuring groundwater-stream water exchange: new techniques for installing minipiezometers and estimating hydraulic conductivity. Trans. Am. Fish. Soc. 132, 493–502. https://doi.org/10.1577/1548-8659(2003)132<0493:MGWENT>2.0.CO;2.

Blanco, N., Polanco, E., 2024. The geological evolution of the Silala River basin, Central Andes. WIREs Water 11 (1), e1695. https://doi.org/10.1002/wat2.1695.

Bower, D., Hannah, D.M., McGregor, G.R., 2004. Techniques for assessing the climatic sensitivity of river flow regimes. Hydrol. Processes 18, 2515–2543. https://doi.org/10.1002/hyp.1479.

Briggs, M.A., Lautz, L.K., Buckley, S.F., Lane, J.W., 2014. Practical limitations on the use of diurnal temperature signals to quantify groundwater upwelling. J. Hydrol. 519, 1739–1751. https://doi.org/10.1016/j.jhydrol.2014.09.030.

Briggs, M.A., Buckley, S.F., Bagtzoglou, A.C., Werkema, D.D., Lane, J.W., 2016. Actively heated high-resolution fiber-optic-distributed temperature sensing to quantify streambed flow dynamics in zones of strong groundwater upwelling. Water Resour. Res. 52, 5179–5194. https://doi.org/10.1002/2015WR018219.

Briggs, M.A., Johnson, Z.C., Snyder, C.D., Hitt, N.P., Kurylyk, B.L., Lautz, L., Irvine, D.J., Hurley, S.T., Lane, J.W., 2018. Inferring watershed hydraulics and cold-water habitat persistence using multi-year air and stream temperature signals. Sci. Total Environ. 636, 1117–1127. https://doi.org/10.1016/j.scitotenv.2018.04.344.

Bristow, K.L., Kluitenberg, G.J., Horton, R., 1994. Measurement of soil thermal properties with a dual-probe heat-pulse technique. Soil Sci. Soc. Am. J. 58 (5), 1288–1294.

Brooks, P.D., Chorover, J., Fan, Y., Godsey, S.E., Maxwell, R.M., McNamara, J.P., Tague, C., 2015. Hydrological partitioning in the critical zone: recent advances and opportunities for developing transferable understanding of water cycle dynamics. Water Resour. Res. 51, 6973–6987. https://doi.org/10.1002/2015WR017039.

Cardenas, M.B., 2010. Thermal skin effect of pipes in streambeds and its implications on groundwater flux estimation using diurnal temperature signals. Water Resour. Res. 46, W03536. https://doi.org/10.1029/2009WR008528.

Carslaw, H.S., Jaeger, J.C., 1959. Conduction of heat in solids. Oxford Univ. Press, Oxford.

Cascarano, R.N., Reeves, D.M., Henry, M.A., 2020. A dye tracer approach for quantifying fluid and solute flux across the sediment-water Interface. Groundwater 59 (3), 428–437. https://doi.org/10.1111/gwat.13060.

Chen, X., Cardenas, M.B., Chen, L., 2015. Three-dimensional versus two-dimensional bed form-induced hyporheic exchange. Water Resources Research 51 (4), 2923–2936. https://doi.org/10.1002/2014WR016848.

Chen, X., Cardenas, M.B., Chen, L., 2018. Hyporheic exchange driven by three-dimensional sandy bed forms: sensitivity to and prediction from bed form geometry. Water Resources Research 54 (6), 4131–4149. https://doi.org/10.1029/2018WR022663.

CM, Vol. 1., 2017. Memorial of the Republic of Chile, I.C.J. Pleadings, Dispute over the Status and Use of the Waters of the Silala (Chile v. Bolivia).

Constantz, J., 2008. Heat as a tracer to determine streambed water exchanges. Water Resour. Res. 44 (4), W00D10 https://doi.org/10.1029/2008WR006996.

Constantz, J., Cox, M.H., Su, G.W., 2003. Comparison of heat and bromide as ground water tracers near streams. Groundwater 41 (5), 647–656. https://doi.org/10.1111/j.1745-6584.2003.tb02403.x.

Cook, P.G., Rodellas, V., Stieglitz, T.C., 2018. Quantifying surface water, porewater, and groundwater interactions using tracers: tracer fluxes, water fluxes, and end-member concentrations. Water Resour. Res. 54 (3), 2452–2465. https://doi.org/10.1002/2017WR021780.

Cuthbert, M.O., Mackay, R., 2013. Impacts of nonuniform flow on estimates of vertical streambed flux. Water Resour. Res. 49 (1), 19–28. https://doi.org/10.1029/2011WR011587.

DIFROL–DICTUC, 2018. SPW-DQN well pumping test, Final Report, May 25th, 2018.

Duque, C., Rosenberry, D.O., 2022. Advances in the study and understanding of groundwater discharge to surface water. Water 14 (11), 1698. https://doi.org/10.3390/w14111698.

Duque, C., Müller, S., Sebok, E., Haider, K., Engesgaard, P., 2016. Estimating groundwater discharge to surface waters using heat as a tracer in low flux environments: the role of thermal conductivity. Hydrol. Process. 30 (3), 383–395. https://doi.org/10.1002/hyp.10568.

Fawcett, T., 2006. An introduction to ROC analysis. Pattern Recognition Letters 27 (8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010.

Ghysels, G., Anibas, C., Awol, H., Tolche, A.D., Schneidewind, U., Huysmans, M., 2021. The significance of vertical and lateral groundwater-surface water exchange fluxes in riverbeds and riverbanks comparing 1D analytical flux estimates with 3D groundwater modelling. Water 13 (3), 306. https://doi.org/10.3390/w13030306.

Gleeson, T., Wagener, T., Döll, P., Zipper, S.C., West, C., Wada, Y., Taylor, R., Scanlon, B., Rosolem, R., Rahman, S., Oshinlaja, N., Maxwell, R., Lo, M., Kim, H., Hill, M., Hartmann, A., Fogg, G., Famiglietti, J.S., Ducharne, A., de Graff, I., Cuthbert, M., Condon, L., Bresciani, E., Bierkens, M.F.P., 2021. GMD perspective: the quest to improve the evaluation of groundwater representation in continental- to global-scale models. Geosci. Model Dev. 14, 7545–7571. https://doi.org/10.5194/gmd-14-7545-2021.

Glose, T.J., Lowry, C.S., Hausner, M.B., 2019. Limits on groundwater-surface water fluxes derived from temperature time series: defining resolution-based thresholds. Water. Resour. Res. 55 (12), 10678–10689. https://doi.org/10.1029/2019WR025643.

Glose, T.J., Lowry, C.S., Hausner, M.B., 2021. Examining the utility of continuously quantified darcy fluxes through the use of periodic temperature time series. J. Hydrol. 595, 125675 https://doi.org/10.1016/j.jhydrol.2020.125675.

Gómez, C., García, S., Suárez, F., Muñoz, J.F., 2024. Hydrogeological characterization of the Silala River catchment. WIREs Water 11 (1), e1697. https://doi.org/10.1002/wat2.1697.

Gordon, R.P., Lautz, L.K., Briggs, M.A., McKenzie, J.M., 2012. Automated calculation of vertical pore-water flux from field temperature time series using the VFLUX method and computer program. J. Hydrol. 420–421, 142–158. https://doi.org/10.1016/j.jhydrol.2011.11.053.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google earth engine: planetary-scale geospatial analysis for everyone. Remote Sens. Environ. 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031.

[dataset] Hall, D.K., Riggs, G.A., 2016. MODIS/Terra Snow Cover Daily L3 Global 500m SIN Grid, Version 6. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MOD10A1.006.

Hansen, P.C., 1992. Analysis of discrete ill-posed problems by means of the L-curve. SIAM Review 34 (4), 561–580. https://doi.org/10.1137/1034115.

Hatch, C.E., Fisher, A.T., Revenaugh, J.S., Constantz, J., Ruehl, C., 2006. Quantifying surface water-groundwater interactions using time series analysis of riverbed thermal records: method development. Water Resour. Res. 42 (10) https://doi.org/10.1029/2005WR004787.

Healy, R.W., Ronan, A.D., 1996. Documentation of computer program VS2DH for simulation of energy transport in variably saturated porous media – modification of the U. In: Geol, U.S., Surv. Water Resour. Invest. (Eds.), S. Geological Survey's Computer Program VS2DT, pp. 96–4230.

Herrera, C., Aravena, R., 2017. Chemical and Isotopic Characterization of the Surface Water and Groundwater of the Silala Transboundary Basin, Second Region, Chile. Memorial of the Republic of Chile, I.C.J. Pleadings, Dispute over the Status and Use of the Waters of the Silala (Chile v. Bolivia). Vol. 4, Annex III.

Hsieh, P.A., Wingle, W., Healy, R.W., 2000. VS2DI - a graphical software package for simulating fluid flow and solute or energy transport in variably saturated porous media: U.S. Geological Survey Water-Resources Investigations Report 99–4130, 16. https://doi.org/10.3133/wri994130.

Humphrey, C.E., Solomon, D.K., Genereux, D.P., Gilmore, T.E., Mittelstet, A.R., Zlotnik, V.A., Zeyrek, C., Jensen, C.R., MacNamara, M.R., 2022. Using automated seepage meters to quantify the spatial variability and net flux of groundwater to a stream. Water Resour. Res. 58 (6), e2021WR030711 https://doi.org/10.1029/2021WR030711.

Irvine, D.J., Cranswick, R.H., Simmons, C.T., Shanafield, M.A., Lautz, L.K., 2015a. The effect of streambed heterogeneity on groundwater-surface water exchange fluxes inferred from temperature time series. Water Resour. Res. 51 (1), 198–212. https://doi.org/10.1002/2014WR015769.

Irvine, D.J., Lautz, L.K., Briggs, M.A., Gordon, R.P., McKenzie, J.M., 2015b. Experimental evaluation of the applicability of phase, amplitude, and combined methods to determine water flux and thermal diffusivity from temperature time series using VFLUX 2. J. Hydrol. 531, 728–737. https://doi.org/10.1016/j.jhydrol.2015.10.054.

Irvine, D.J., Briggs, M.A., Lautz, L.K., Gordon, R.P., McKenzie, J.M., Cartwright, I., 2017. Using diurnal temperature signals to infer vertical groundwater-surface water exchange. Groundwater 55 (1), 10–26. https://doi.org/10.1111/gwat.12459.

Johnson, D.O., Arriaga, F.J., Lowery, B., 2005. Automation of a falling head permeameter for rapid determination of hydraulic conductivity of multiple samples. Soil Sci. Soc. Am. J. 69 (3), 828–833. https://doi.org/10.2136/sssaj2004.0014N.

Kalbus, E., Reinstorf, F., Schirmer, M., 2006. Measuring methods for groundwater – surface water interactions: a review. Hydrol. Earth Syst. Sci. Discuss. 10 (6), 873–887. https://doi.org/10.5194/hess-10-873-2006.

Keery, J., Binley, A., Crook, N., Smith, J.W.N., 2007. Temporal and spatial variability of groundwater-surface water fluxes: development and application of an analytical method using temperature time series. J. Hydrol. 336 (1–2), 1–16. https://doi.org/10.1016/j.jhydrol.2006.12.003.

Klaus, J., McDonnell, J.J., 2013. Hydrograph separation using stable isotopes: review and evaluation. J. Hydrol. 505 (2013), 47–64. https://doi.org/10.1016/j.jhydrol.2013.09.006.

Koch, F.W., Voytek, E.B., Day-Lewis, F.D., Healy, R., Briggs, M.A., Lane Jr., J.W., Werkema, D., 2015. 1DTempPro V2: new features for inferring groundwater/surface-water exchange. Groundwater 54 (3). https://doi.org/10.1111/gwat.12369.

Kurylyk, B.L., Irvine, D.J., Bense, V.F., 2019. Theory, tools, and multidisciplinary applications for tracing groundwater fluxes from temperature profiles. Wires Water 6 (1), e1329.

Latorre, C., Frugone, M., 2017. Annex IV – Holocene Sedimentary History of the Río Silala (Antofagasta Region, Chile). In Dispute over the Status and Use of the Waters of the Silala (Chile v. Bolivia) Annexes IV – X To The Experts Reports, (Vol. 5, pp. 1–38). I.C.J. https://www.icj-cij.org/public/files/case-related/162/162-20170703-WRI-01-04-EN.pdf.

Lagos, M.S., Muñoz, J.F., Suárez, F., Fuenzalida, M.J., Yáñez-Morroni, G., Sanzana, P., 2024. Investigating the effects of channelization in the Silala River: A review of the implementation of a coupled MIKE-11 and MIKE-SHE modeling system. WIREs Water 11 (1), e1673. https://doi.org/10.1002/wat2.1673.

Latorre, C., Frugone-Álvarez, M., 2024. Holocene sedimentary history of the Silala River (Antofagasta region, Chile). WIREs Water 11 (1), e1699. https://doi.org/10.1002/wat2.1699.

Lautz, L.K., 2010. Impacts of nonideal field conditions on vertical water velocity estimates from streambed temperature time series. Water Resour. Res. 46 (1) https://doi.org/10.1029/2009WR007917.

Lee, D.R., Cherry, J.A., 1979. A field exercise on groundwater flow using seepage meters and mini-piezometers. J. Geol. Educ. 27 (1), 6–10. https://doi.org/10.5408/0022-1368-27.1.6.

Lettenmaier, D.P., Wood, A.W., Palmer, R.N., Wood, E.F., Stakhiv, E.Z., 1999. Water resources implications of global warming: a U.S. regional perspective. Clim. Change 43 (3), 537–579. https://doi.org/10.1023/A:1005448007910.

Luce, C.H., Tarboton, D.G., 2010. Evaluation of alternative formulae for calculation of surface temperature in snowmelt models using frequency analysis of temperature observations. Hydrol. Earth Syst. Sci. 14 (3), 535–543. https://doi.org/10.5194/hess-14-535-2010.

Luce, C.H., Tonina, D., Gariglio, F., Applebee, R., 2013. Solutions for the diurnally forced advection-diffusion equation to estimate bulk fluid velocity and diffusivity in streambeds from temperature time series. Water Resour. Res. 49 (1), 488–506. https://doi.org/10.1029/2012WR012380.

Ma, Z., Wang, W., Zhao, M., Hou, X., Wang, Z., Zheng, H., 2022. Non-darcy flow through a natural streambed in a disconnected stream. Water Resour. Res. 58 (3), e2021WR031356 https://doi.org/10.1029/2021WR031356.

Mao, L., 2024. The geomorphological and ecological functioning of the Silala River. WIREs Water 11 (1), e1691. https://doi.org/10.1002/wat2.1691.

McAliley, W.A., Day-Lewis, F.D., Rey, D., Briggs, M.A., Shapiro, A.M., Werkema, D., 2022. Application of recursive estimation to heat tracing for groundwater/surface-water exchange. Water Resour. Res. 58 (6), e2021WR030443 https://doi.org/10.1029/2021WR030443.

McCallum, A.M., Andersen, M.S., Rau, G.C., Acworth, R.I., 2012. A 1-D analytical method for estimating surface water-groundwater interactions and effective thermal diffusivity using temperature time series. Water Resour. Res. 48 (11), W11532 https://doi.org/10.1029/2012WR012007.

Muñoz, J.F., Suárez, F., Alcayaga, H., McRostie, V., Fernández, B., 2024. Introduction to the Silala. River and its hydrology. WIREs Water 11 (1), e1644. https://doi.org/10.1002/wat2.1644.

Muñoz, J.F., Suárez, F., Fernández, B., Maass, T., 2017. Annex VII – hydrology of the Silala River basin. in dispute over the status and use of the waters of the silala (Chile v. I.C.J Bolivia) Annexes IV – X to the Experts Reports 5, 163–278. https://www.icj-cij.org/public/files/case-related/162/162-20170703-WRI-01-04-EN.pdf.

Munz, M., Schmidt, C., 2017. Estimation of vertical water fluxes from temperature time series by the inverse numerical computer program FLUX-BOT. Hydrol. Processes 31 (15), 2713–2724. https://doi.org/10.1002/hyp.11198.

Munz, M., Oswald, S.E., Schmidt, C., 2011. Sand box experiments to evaluate the influence of subsurface temperature probe design on temperature based water flux calculation. Hydrol. Earth Syst. Sci. 15 (11), 3495–3510. https://doi.org/10.5194/hess-15-3495-2011.

Naranjo, R., Smith, D., Lindenbach, E., 2023. Incorporating temperature into seepage loss estimates for a large unlined irrigation canal. J. Hydrol. 617, 129117 https://doi.org/10.1016/j.jhydrol.2023.129117.

Naranjo, R.C., Turcotte, R., 2015. A new temperature profiling probe for investigating groundwater-surface water interaction. Water Resour. Res. 51 (9), 7790–7797. https://doi.org/10.1002/2015WR017574.

Niswonger, R.G., Prudic, D.E., 2003. Modeling heat as a tracer to estimate streambed seepage and hydraulic conductivity, in: Constantz, J., Stonestrom, D.A. (Eds.), Heat as a Tool for Studying the Movement of Ground Water Near Streams. U.S. Geol. Surv. Circ. 1260, pp. 81–89. https://doi.org/10.3133/cir1260.

Panagoulia, D., Dimou, G., 1996. Sensitivities of groundwater-stream-flow interaction to global climate change. Hydrol. Sci. J. 41 (5), 781–796. https://doi.org/10.1080/02626669609491545.

Peach, D., Taylor, A., 2024. The development of a hydrogeological conceptual model of groundwater and surface water flows in the Silala River basin. WIREs Water 11 (1), e1676. https://doi.org/10.1002/wat2.1676.

Rau, G.C., Andersen, M.S., Acworth, R.I., 2012. Experimental investigation of the thermal dispersivity term and its significance in the heat transport equation for flow in sediments. Water Resour. Res. 48 (3), W03511 https://doi.org/10.1029/2011WR011038.

Ren, J., Zhang, J., Wang, D., 2023. Validation of an analytical model of groundwater velocity based on laboratory test and numerical simulation. Environ. Earth Sci. 82 (10), 258. https://doi.org/10.1007/s12665-023-10959-3.

Rieu-Clarke, A., Moynihan, R., Magsig, B.-O., 2012. UN watercourses convention: user's guide. IHP-HELP Centre for Water Law, Policy and Science (under the auspices of UNESCO).

Rosenberry, D.O., Briggs, M.A., Delin, G., Hare, D.K., 2016. Combined use of thermal methods and seepage meters to efficiently locate, quantify, and monitor focused groundwater discharge to a sand-bed stream. Water Resour. Res. 52 (6), 4486–4503. https://doi.org/10.1002/2016WR018808.

Rosenberry, D.O., Engesgaard, P., Hatch, C., 2021. Hydraulic conductivity can no longer be considered a fixed property when quantifying flow between groundwater and surface water. Hydrol. Processes 35 (6). https://doi.org/10.1002/hyp.14226.

Ruehl, C., Fisher, A.T., Hatch, C., Los Huertos, M., Stemler, G., Shennan, C., 2006. Differential gauging and tracer tests resolve seepage fluxes in a strongly-losing stream. J. Hydrol. 330 (1–2), 235–248. https://doi.org/10.1016/j.jhydrol.2006.03.025.

Saar, M.O., 2011. Geothermal heat as a tracer of large-scale groundwater flow and as a means to determine permeability fields. Hydrogeol. J. 19 (1), 31–52. https://doi.org/10.1007/s10040-010-0657-2.

Särkkä, S., 2013. Bayesian filtering and smoothing. Cambridge Univ. Press, Cambridge.

Scanlon, B.R., Fakhreddine, S., Rateb, A., de Graaf, I., Famiglietti, J., Gleeson, T., Grafton, R.Q., Jobbagy, E., Kebede, S., Kolusu, S.R., Konikow, L.F., Long, D., Mekonnen, M., Schmied, H.M., Mukherjee, A., MacDonald, A., Reedy, R.C., Shamsudduha, M., Simmons, C.T., Sun, A., Taylor, R.G., Villholth, K.G., Vörösmarty, C.J., Zheng, C., 2023. Global water resources and the role of groundwater in a resilient water future. Nat. Rev. Earth Environ. 4 (2), 87–101. https://doi.org/10.1038/s43017-022-00378-6.

Schneidewind, U., van Berkel, M., Anibas, C., Vandersteen, G., Schmidt, C., Joris, I., Seuntjens, P., Batelaan, O., Zwart, H.J., 2016. LPMLE3: a novel 1-D approach to study water flow in streambeds using heat as a tracer. Water Resour. Res. 52 (8), 6596–6610. https://doi.org/10.1002/2015WR017453.

Shanafield, M., Pohll, G., Susfalk, R., 2010. Use of heat-based vertical fluxes to approximate total flux in simple channels. Water Resour. Res. 46 (3), W03508 https://doi.org/10.1029/2009WR007956.

Shi, W., Zhan, H., Wang, Q., 2023. Quantifying vertical streambed fluxes and streambed thermal properties using heat as a tracer during extreme hydrologic events. J. Hydrol. 130553 https://doi.org/10.1016/j.jhydrol.2023.130553.

Simon, N., Bour, O., Faucheux, M., Lavenant, N., Le Lay, H., Fovet, O., Thomas, Z., Longuevergne, L., 2022. Combining passive and active distributed temperature sensing measurements to locate and quantify groundwater discharge variability into a headwater stream. Hydrol. Earth Syst. Sci. 26 (5), 1459–1479. https://doi.org/10.5194/hess-26-1459-2022.

Simon, N., Bour, O., Lavenant, N., Porel, G., Nauleau, B., Klepikova, M., 2023. Monitoring groundwater fluxes variations through active-DTS measurements. J. Hydrol. 622, 129755 https://doi.org/10.1016/j.jhydrol.2023.129755.

Singh, D., Singh, B., 2020. Investigating the impact of data normalization on classification performance. Appl. Soft Comput. 97, 105524 https://doi.org/10.1016/j.asoc.2019.105524.

Smith, L., Schwartz, F.W., 1981. Mass transport: 3. role of hydraulic conductivity data in prediction. Water Resour. Res. 17 (5), 1463–1479. https://doi.org/10.1029/WR017i005p01463.

Sophocleous, M., 2004. Climate change: why should water professionals care? Groundwater 42 (5), 637–638. https://doi.org/10.1111/j.1745-6584.2004.tb02715.x.

Stallman, R.W., 1965. Steady one-dimensional fluid flow in a semi-infinite porous medium with sinusoidal surface temperature. J. Geophys. Res. 70 (12), 2821–2827. https://doi.org/10.1029/JZ070i012p02821.

Suárez, F., Sandoval, V., Sarabia, A., 2017. Annex X – river-aquifer interactions using heat as a tracer in the Transboundary Basin of the Silala River. in dispute over the status and use of the waters of the silala (Chile v. I.C.J Bolivia) Annexes IV – X to the Expert Reports 5, 475–576. https://www.icj-cij.org/public/files/case-related/162/162-20170703-WRI-01-04-EN.pdf.

Suárez, F., Sandoval, V., Sarabia, A., Muñoz, J.F., 2024a. Investigating river-aquifer interactions using heat as a tracer in the silala river transboundary basin. WIREs Water 11 (1), e1639. https://doi.org/10.1002/wat2.1639.

Suárez, F., Oportus, T., Mendoza, M., Aguirre, I., Godoy, V., Muñoz, J.F., 2024b. Evaporation processes in the Silala River basin. WIREs Water 11 (1), e1638. https://doi.org/10.1002/wat2.1638.

Suárez, F., Latorre, C., Mendoza, M., Frugone, M., Muñoz, J.F., 2024c. Using foliar δ13C from high-andean plants (Silala River basin) as a measure of potential evapotranspiration through water use efficiency. WIREs Water 11 (1), e1707. https://doi.org/10.1002/wat2.1707.

Suárez, F., Sarabia, A., Sanzana, P., Latorre, C., Muñoz, J.F., 2024d. The quebrada negra wetland study: an approach to understand plant diversity, hydrology, and hydrogeology of high-andean wetlands. WIREs Water 11 (1), e1683. https://doi.org/10.1002/wat2.1683.

Taylor, A., Peach, D., 2024. Groundwater modeling of the silala basin and impacts of channelization. WIREs Water 11 (1), e1662. https://doi.org/10.1002/wat2.1662.

Taylor, C.J., Pedregal, D.J., Young, P.C., Tych, W., 2007. Environmental time series analysis and forecasting with the captain toolbox. Environ. Model. Softw. 22 (6), 797–814. https://doi.org/10.1016/j.envsoft.2006.03.002.

van Kampen, R., Schneidewind, U., Anibas, C., Bertagnoli, A., Tonina, D., Vandersteen, G., Luce, C., Krause, S., van Berkel, M., 2022. LPMLEn-A frequency domain method to estimate vertical streambed fluxes and sediment thermal properties in semi-infinite and bounded domains. Water Resour. Res. 58 (3), e2021WR030886 https://doi.org/10.1029/2021WR030886.

Vandersteen, G., Schneidewind, U., Anibas, C., Schmidt, C., Seuntjens, P., Batelaan, O., 2015. Determining groundwater-surface water exchange from temperature-time series: combining a local polynomial method with a maximum likelihood estimator. Water Resour. Res. 51 (2), 922–939. https://doi.org/10.1002/2014WR015994.

Wheater, H.S., Peach, D.W., Suárez, F., Muñoz, J.F., 2024. Understanding the Silala River – scientific insights from the dispute over the status and use of the waters of the silala (Chile v. Bolivia). WIREs Water 11 (1), e1663. https://doi.org/10.1002/wat2.1663.

Wörman, A., Riml, J., Schmadel, N., Neilson, B.T., Bottacin-Busolin, A., Heavilin, J.E., 2012. Spectral scaling of heat fluxes in streambed sediments. Geophys. Res. Lett. 39 (23), L23402 https://doi.org/10.1029/2012GL053922.

Yañez-Morroni, G., Suárez, F., Muñoz, J.F., Lagos, M., 2024. Hydrogeological modelling of the Silala River basin. 2. validation of hydrological fluxes with contemporary data. WIREs Water 11 (1), e1696. https://doi.org/10.1002/wat2.1696.

Young, P.C., Pedregal, D.J., Tych, W., 1999. Dynamic harmonic regression. J. Forecast. 18 (6), 369–394. https://doi.org/10.1002/(SICI)1099-131X(199911)18:6<369::AID-FOR748>3.0.CO;2-K.

Zhang, H., Zhang, F., Zhang, G., Che, T., Yan, W., Ye, M., Ma, N., 2019. Ground-based evaluation of MODIS snow cover product V6 across China: implications for the selection of NDSI threshold. Sci. Total Environ. 651, 2712–2726. https://doi.org/10.1016/j.scitotenv.2018.10.128.